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High Blood Pressure:

Hypertension

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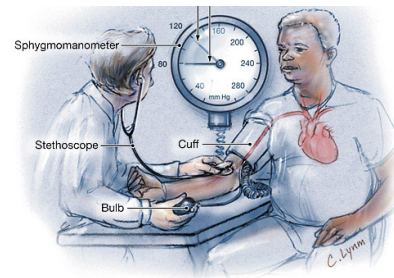
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2022

What is Hypertension (HT)?

- HT is the presence of abnormally $\uparrow$ BP ($>140/90$ mmHg)
- Progressive “**CV syndrome**” associated with target organs damage, often present before high BP values are observed
- Elevated diastolic BP as well as systolic BP are equally important: risk of $\uparrow 20$ mmHg systolic BP = $\uparrow 10$ mmHg diastolic BP
- Raised systolic BP without significantly raised diastolic BP is termed “**Isolated Systolic Hypertension (ISH)**” - common in elderly



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Burden of Hypertension in Asia

Hypertension affects approximately 1 billion people worldwide

- 2/3 in developing countries
- 1/3 adult in South-East Asia has hypertension
- 1.5 million people die of hypertension related diseases each year in South-East Asia

China¹
27.2%

Korea³
33.7%

Taiwan⁶
25%

Philippines⁵
17.4%

Hong Kong²
20%

Malaysia⁴
33%

Thailand⁷
21%

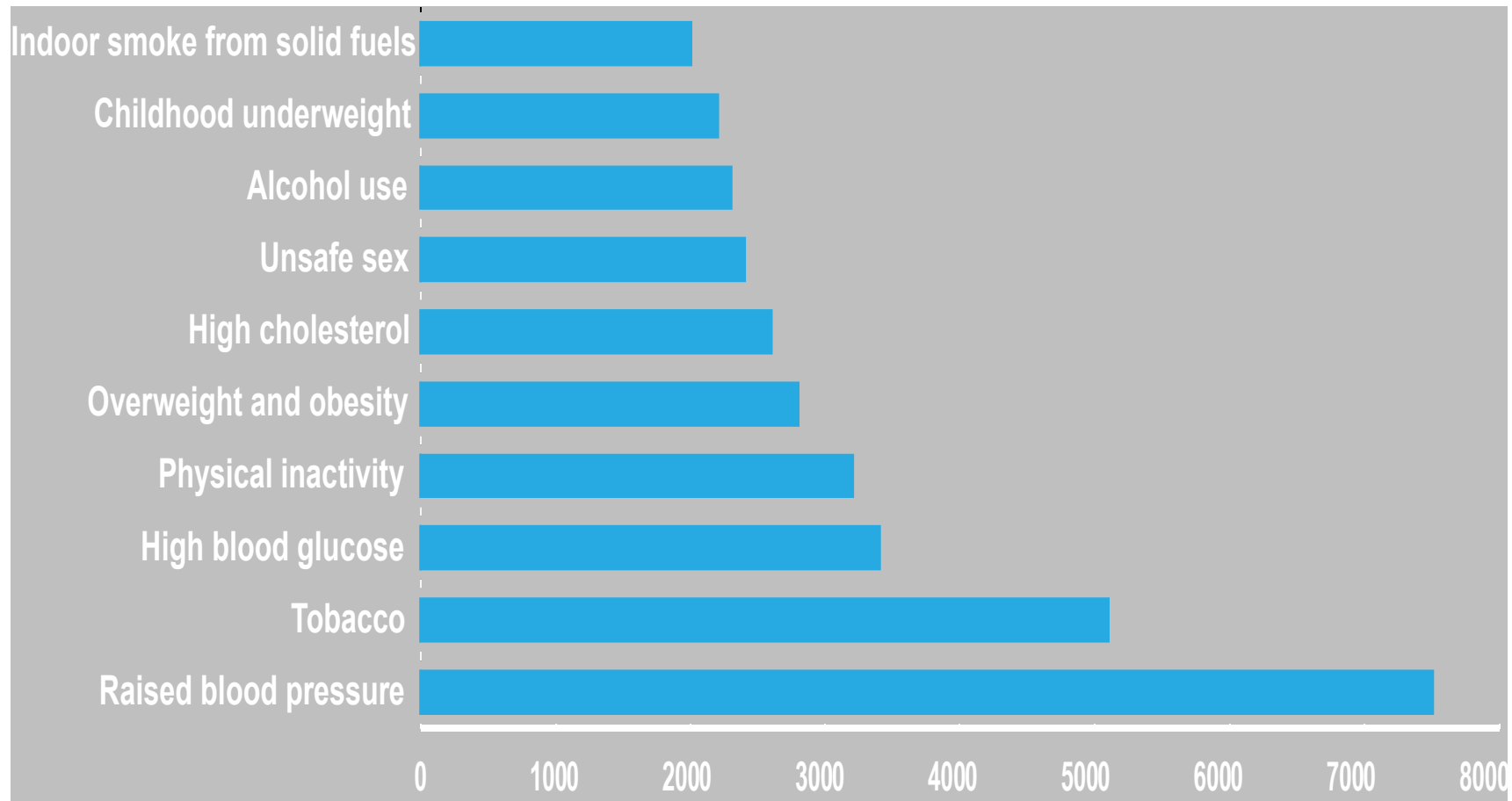


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1. Gu FD, et al. Hypertens 2002;40:920-927. 2. Cheung BM. Hong Kong Med J 2003;9(3):224-226. 3. JoL, et al J Hypertens 2001; 19(9):1523-1532. 4. Lim TO, et al. Singapore med J 2004; 45 (1):20-27. 5. Dans AL, et al. Phil J Internal Med 2005;43:103-115. 6. Fujita T, et al. Int J Clin Pract 2006;60(Suppl 150):3-6. 7. Singh RB, et al J Human Hypertens 2000;14:749-763

Ranking of Top 10 Risk Factors for Death Worldwide

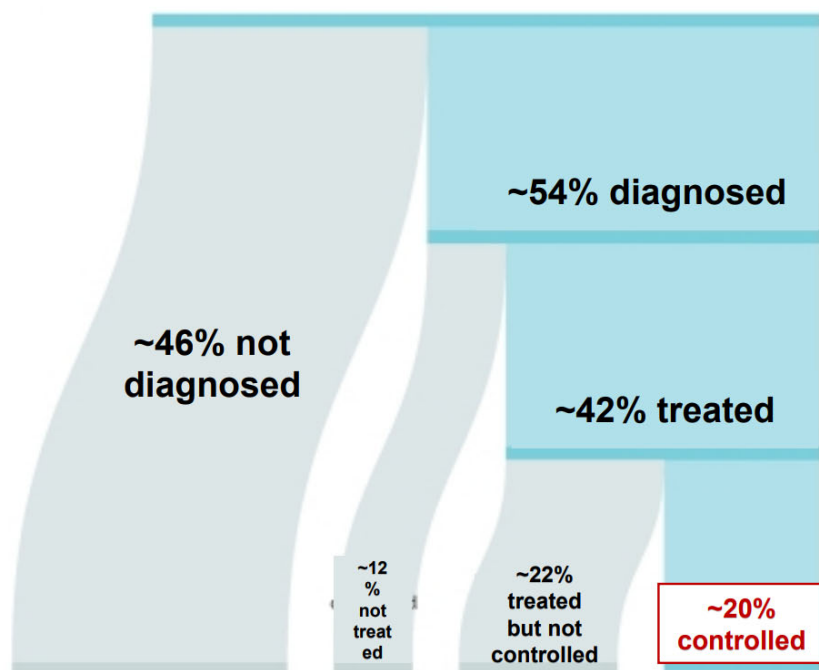


Attributable deaths due to selected risk factors (in thousands)

Global Trends in Diagnosis, Treatment and Control of Hypertension

- The control of hypertension remains largely unsatisfactory worldwide

NCD-RisC 2021



A pooled analysis of 1201 population-representative studies with 104 million participants from 1990 to 2019

China Hypertension Survey, 2012–2015

	Controlled Rate (%)	
	<140/90 mmHg	<130/80 mmHg
Overall	<u>15.3</u>	<u>3.0</u>
Age Groups (years)		
18-24	0.6	0.0
25-34	3.2	0.2
35-44	9.9	1.2
45-54	16.1	2.9
55-64	18.6	5.0
65-74	18.4	5.9
>75	17.0	5.4

Heavy burden of CVD and related health care costs

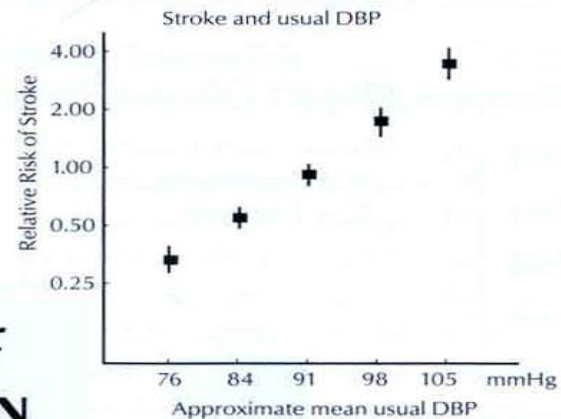
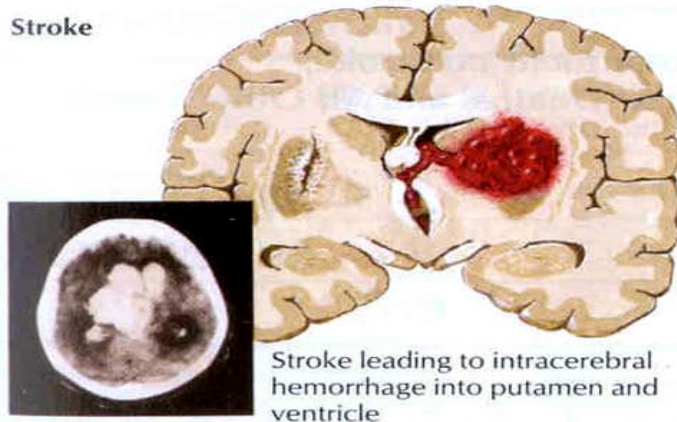
Lancet. 2021; 398(10304):957-980
Circulation. 2018;137(22):2344-235



HT as Risk Factor for CVD

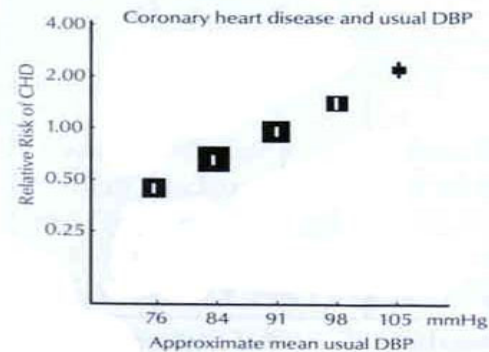
Level of blood pressure is associated with cardiovascular events in a continuous, graded, and apparently independent fashion**

Stroke

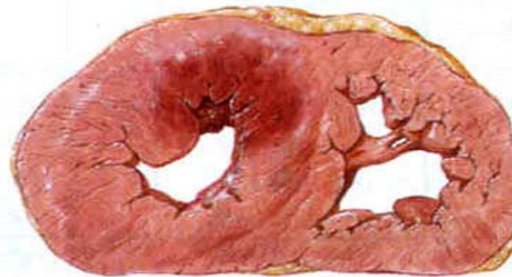


F. Netter M.D.
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LEARNING SYSTEMS

Coronary Heart Disease



Cardiac hypertrophy and antero-septal infarct with coronary heart disease



Angina

** Relative risk of stroke and coronary heart disease as a function of usual diastolic pressure in 420,000 individuals 25 years or older with a mean follow-up period of 10 years. Adapted from MacMahon S, Peto R, Cutler S, et al. Blood pressure, stroke, and coronary heart disease: part one. *Lancet*. 1990;335:765-774.

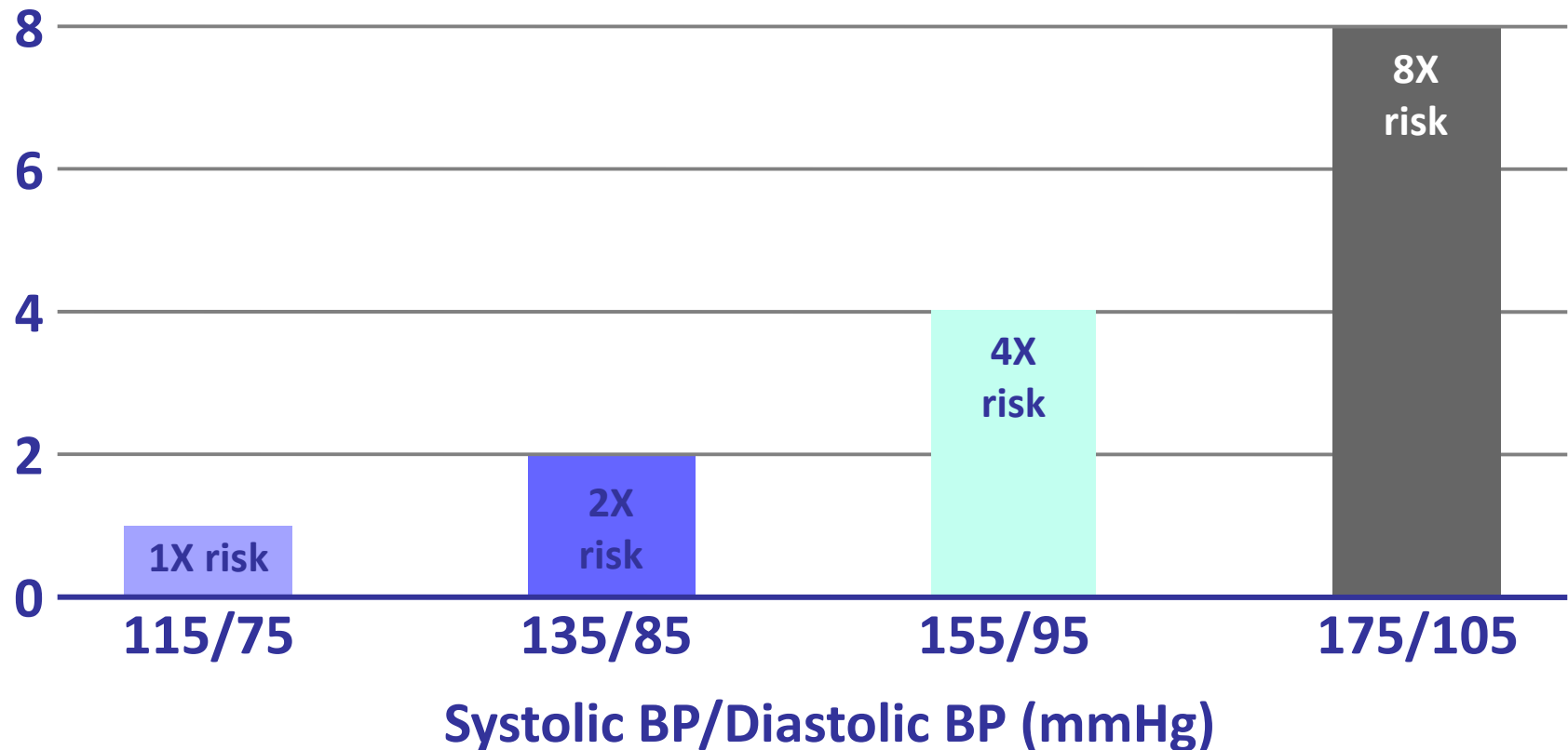


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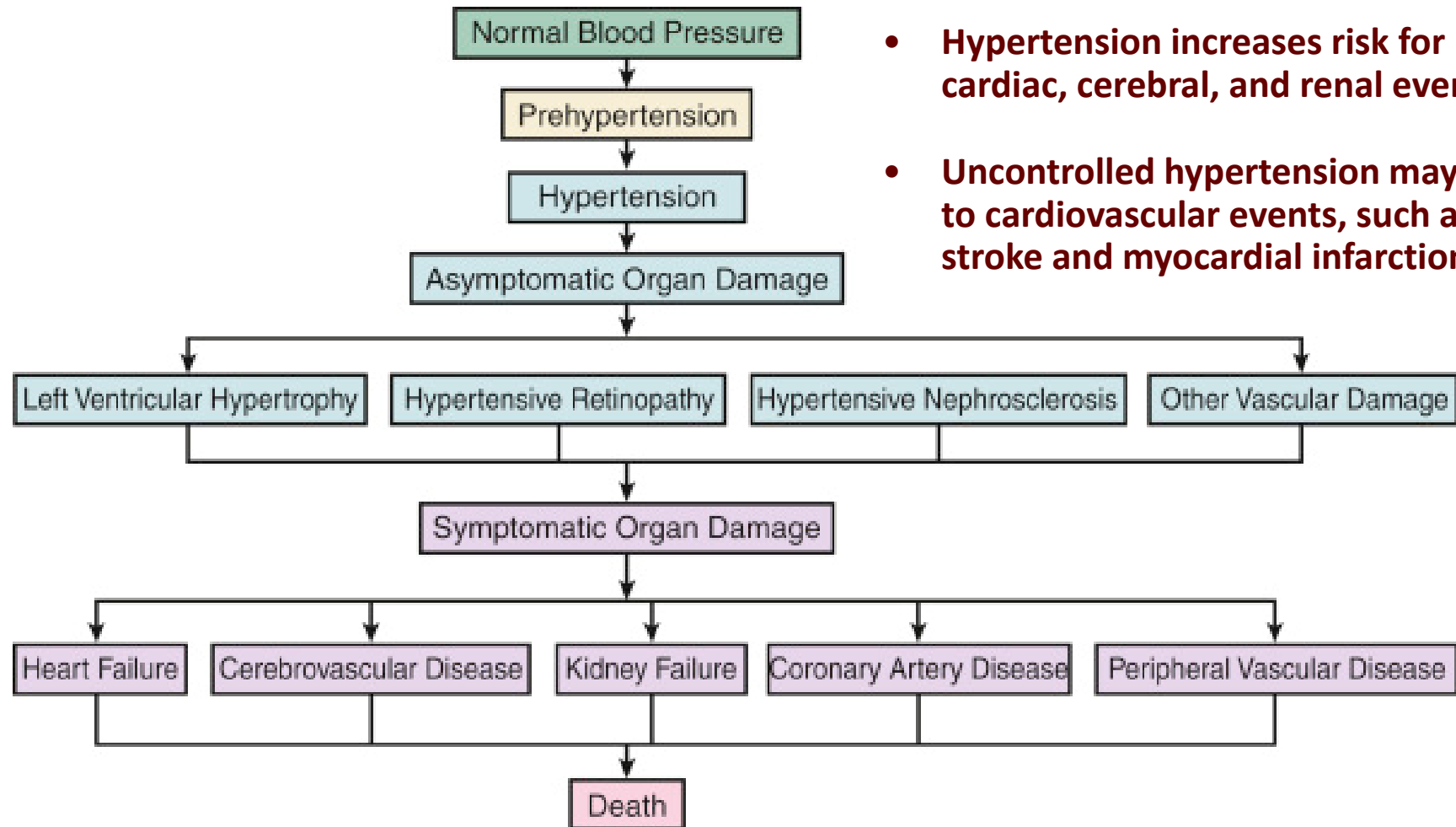
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Cardiovascular Mortality Risk Doubles with each 20/10 mmHg Increase in Systolic/Diastolic BP*

Cardiovascular mortality risk



Uncontrolled Hypertension and Risk for Cardiovascular Events

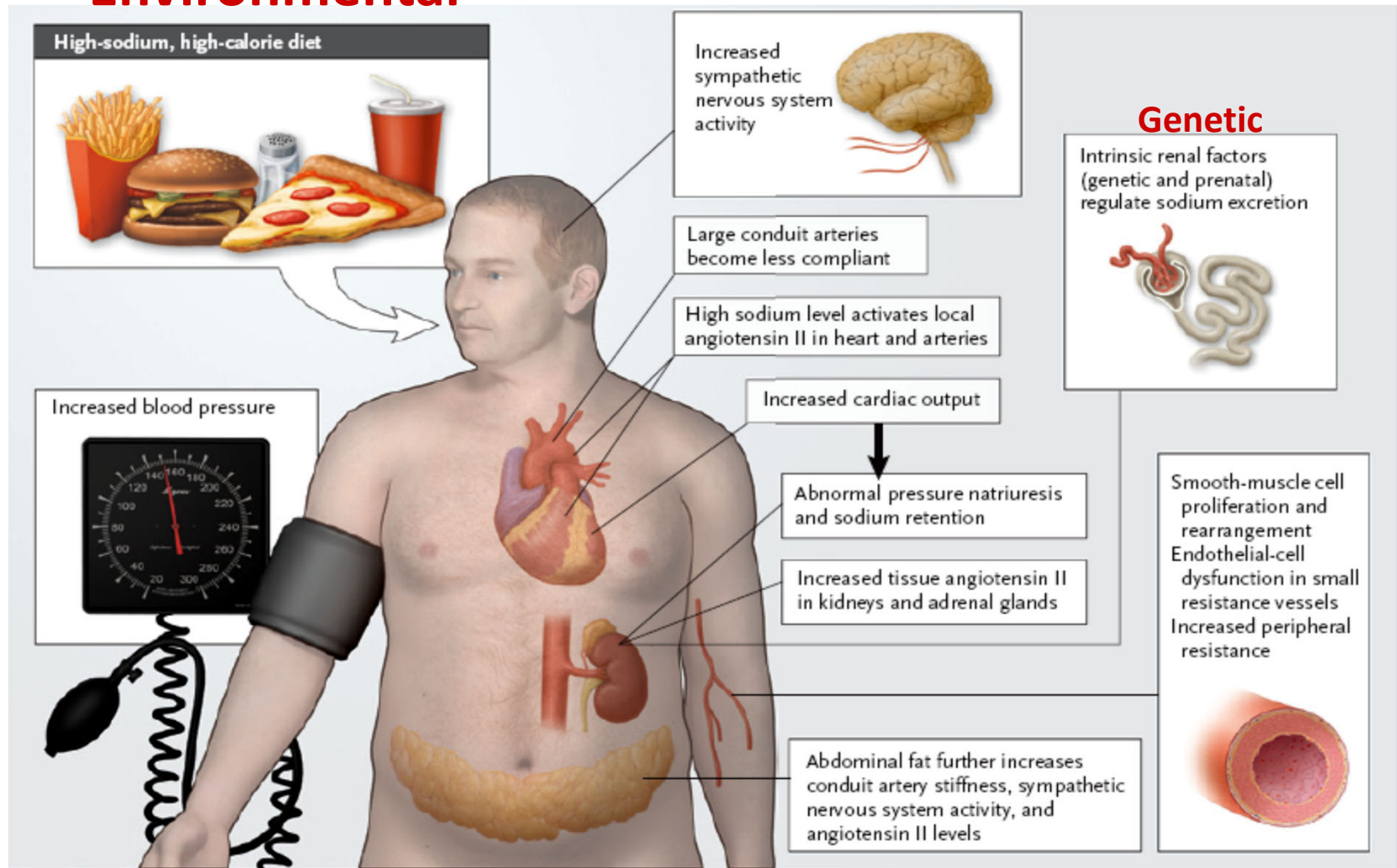


- Hypertension increases risk for cardiac, cerebral, and renal events
- Uncontrolled hypertension may lead to cardiovascular events, such as stroke and myocardial infarction

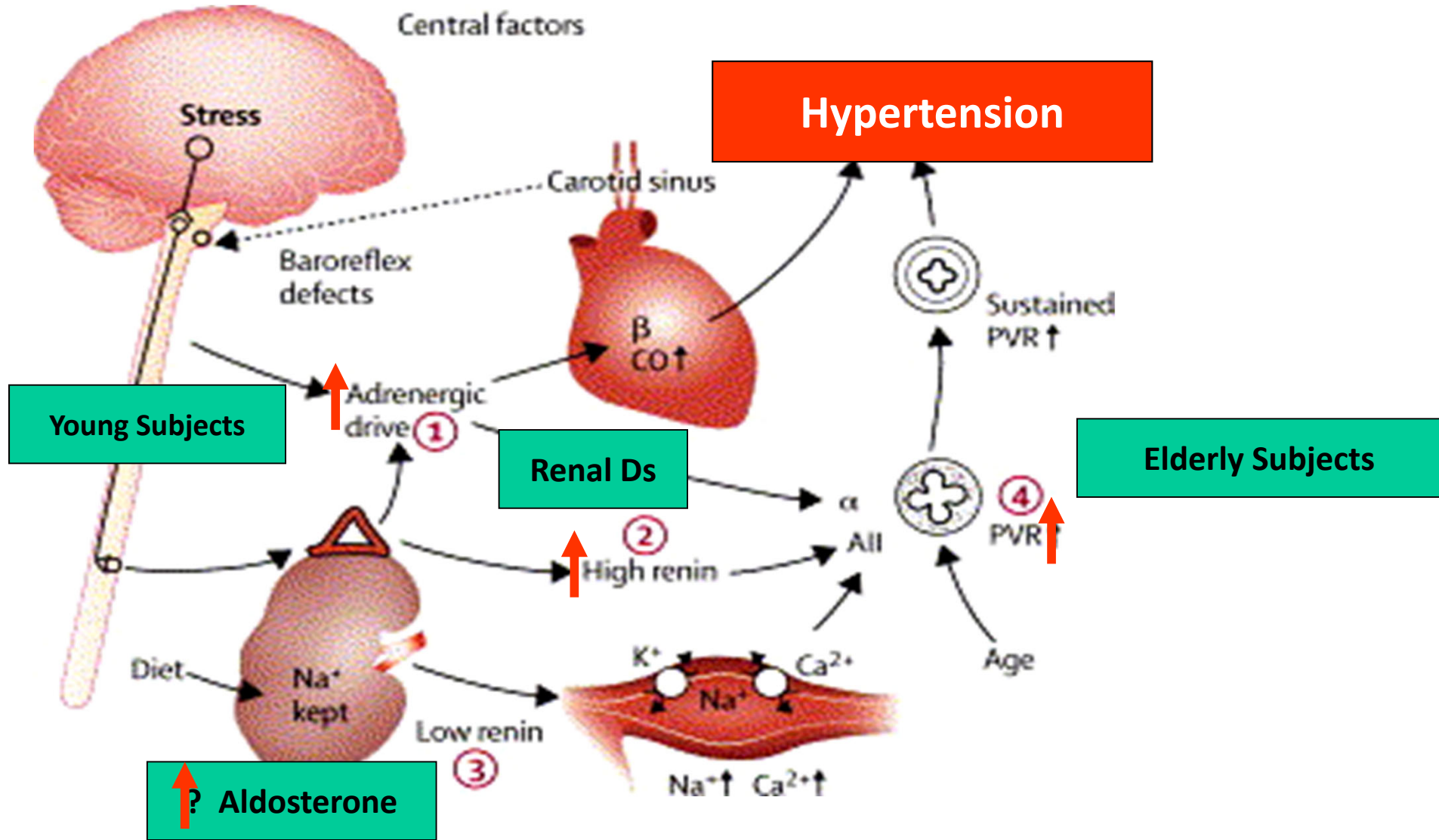


What Cause “Essential” HT?

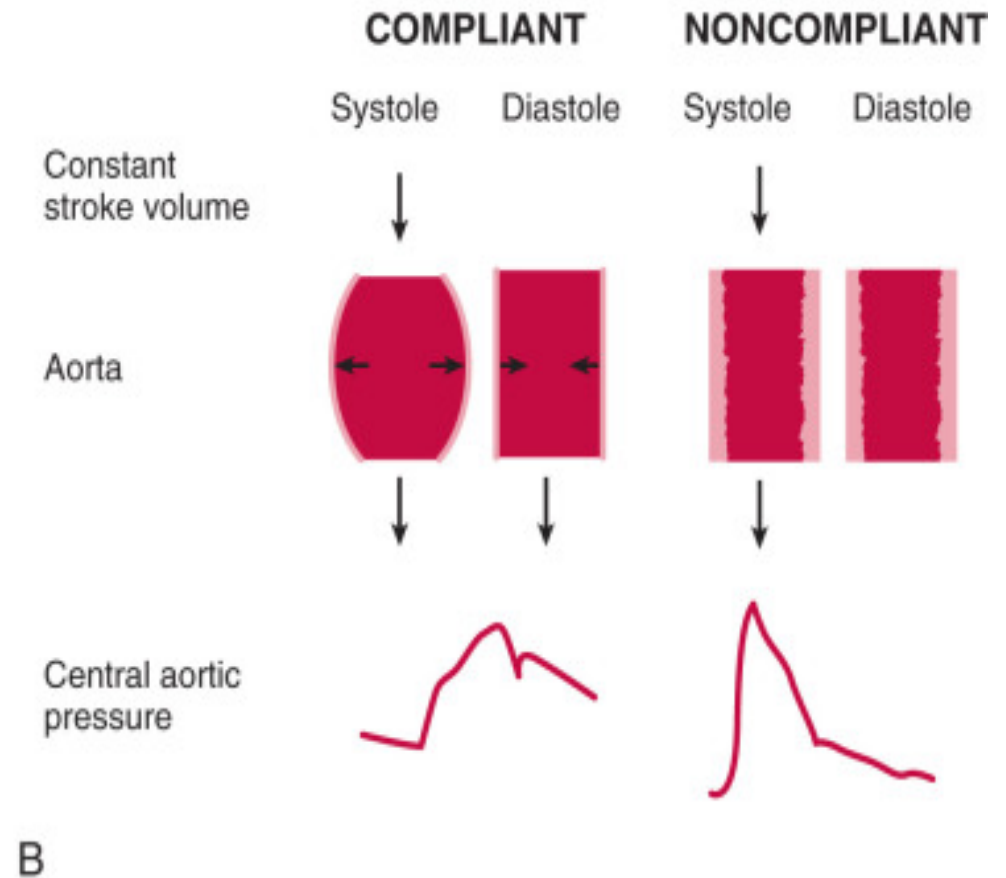
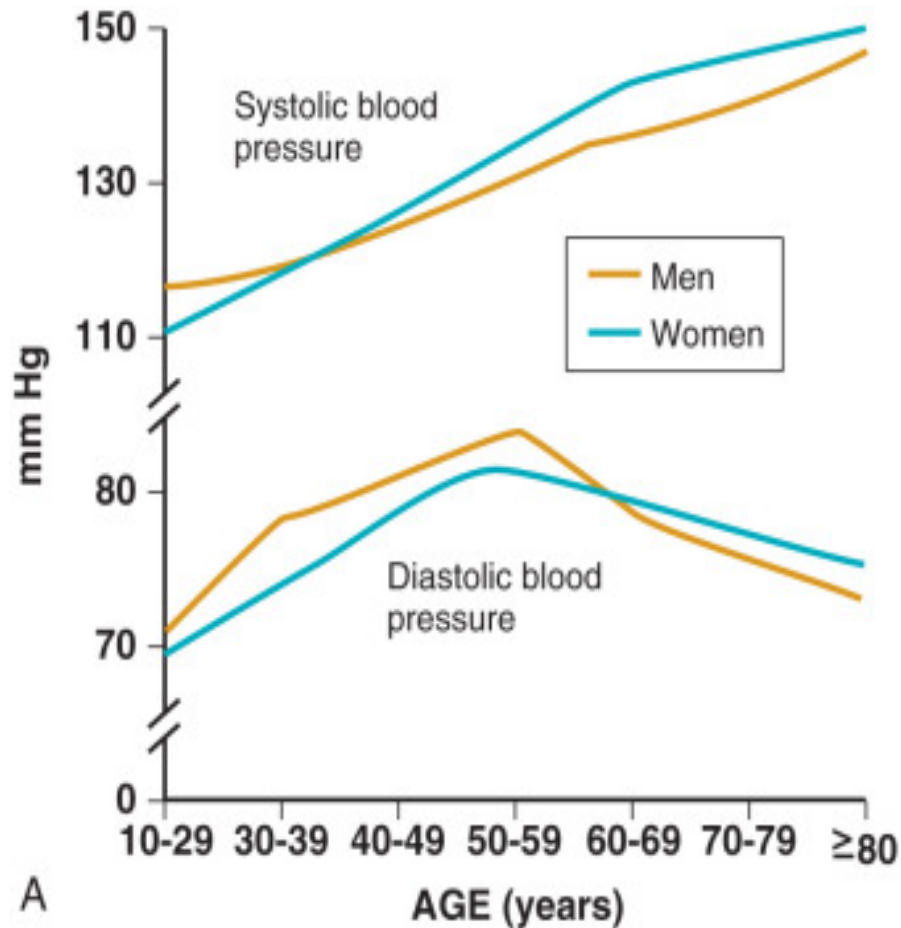
Environmental



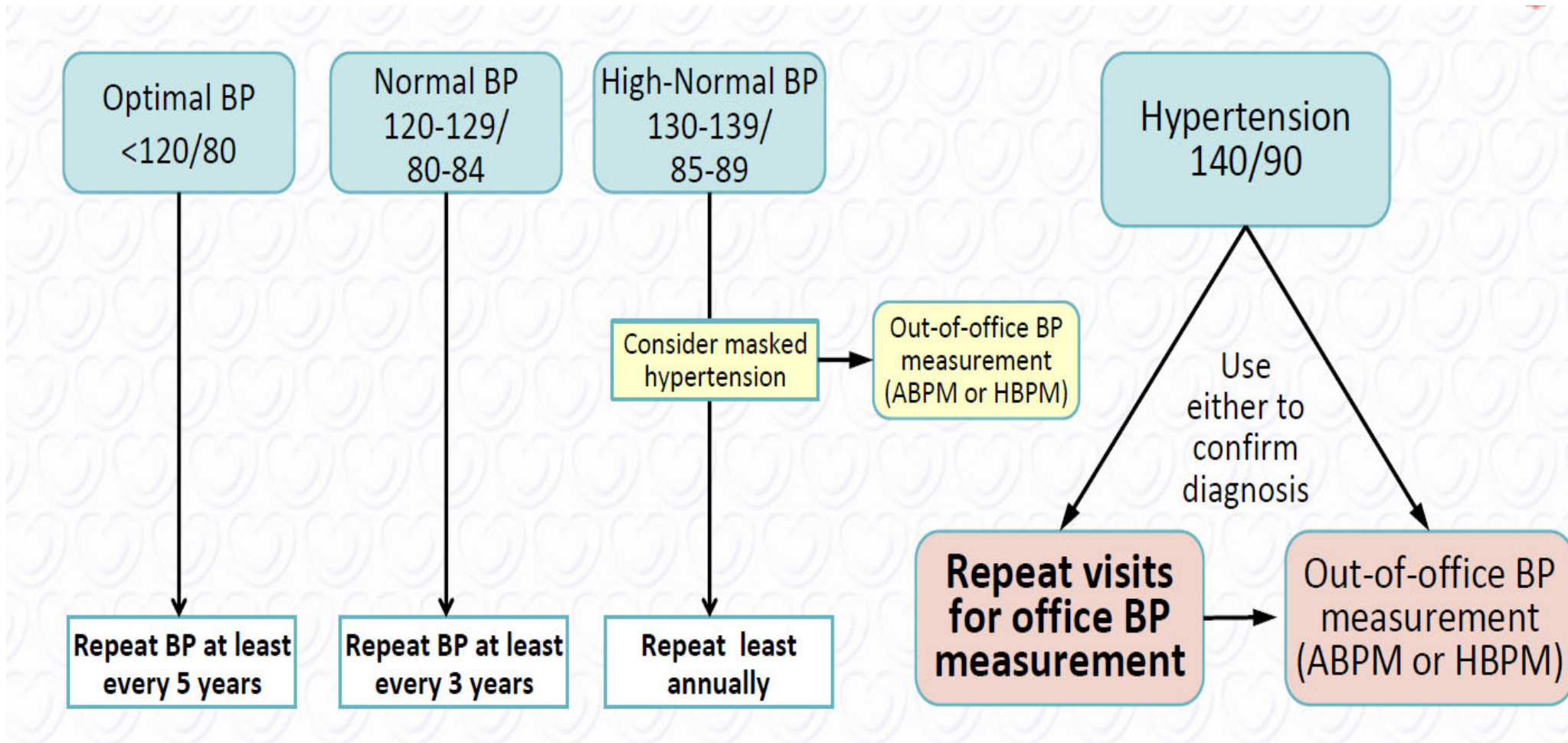
Mechanisms of Hypertension



Effects of Aging on Hypertension



How should hypertension be diagnosed?



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European Society of Cardiology 2018

Measurement of BP

Conditions	• Quiet room with comfortable temperature.
	• Before measurements: Avoid smoking, caffeine and exercise for 30 min; empty bladder; remain seated and relaxed for 3–5 min.
	• Neither patient nor staff should talk before, during and between measurements.
Positions	• Sitting: Arm resting on table with mid-arm at heart level; back supported on chair; legs uncrossed and feet flat on floor (Figure 1).
Device	• Validated electronic (oscillometric) upper-arm cuff device. Lists of accurate electronic devices for office, home and ambulatory BP measurement in adults, children and pregnant women are available at www.stridebp.org.²² (see also Section 11: Resources)
	• Alternatively use a calibrated auscultatory device, (aneroid, or hybrid as mercury sphygmomanometers are banned in most countries) with 1st Korotkoff sound for systolic blood pressure and 5th for diastolic with a low deflation rate.²²
Cuff	• Size according to the individual's arm circumference (smaller cuff overestimates and larger cuff underestimates blood pressure).
	• For manual auscultatory devices the inflatable bladder of the cuff must cover 75%–100% of the individual's arm circumference. For electronic devices use cuffs according to device instructions.
Protocol	• At each visit take 3 measurements with 1 min between them. Calculate the average of the last 2 measurements. If BP of first reading is <130/85 mm Hg no further measurement is required.
Interpretation	• Blood pressure of 2–3 office visits $\geq 140/90$ mm Hg indicates hypertension.



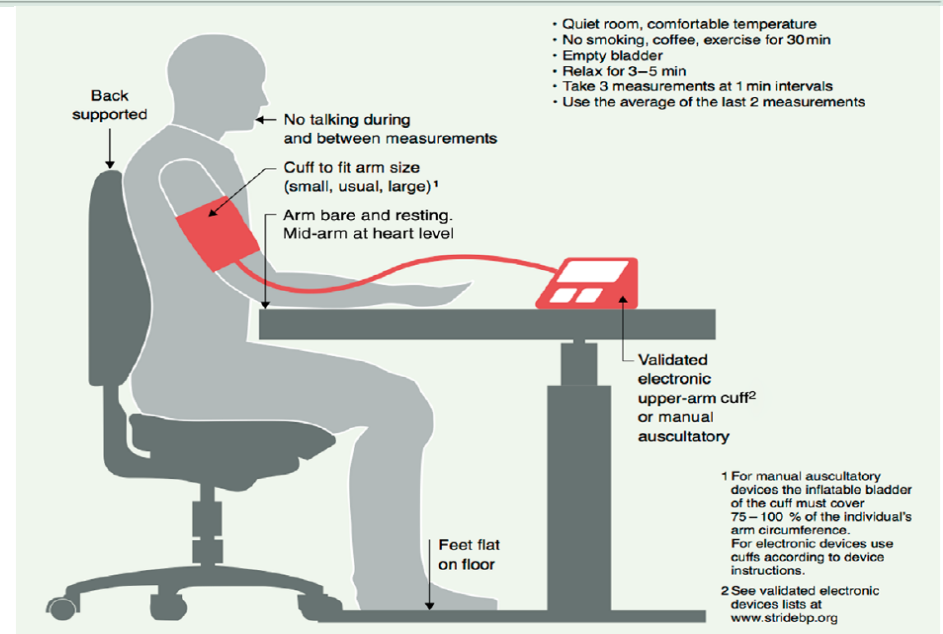
Definition for hypertension

	Systolic and diastolic blood pressure, mm Hg
American College of Cardiology–American Heart Association⁴⁹	
Normal	<120 and <80
Increased	120–129 and <80
Stage 1	130–139 or 80–89
Stage 2	≥140 or ≥90
European Society of Cardiology–European Society of Hypertension⁴⁸	
Optimal	<120 and <80
Normal	120–129 or 80–84, or both
High-normal	130–139 or 85–89, or both
Stage 1	140–159 or 90–99, or both
Stage 2	160–179 or 100–109, or both
Stage 3	≥180 or ≥110, or both
Isolated systolic hypertension	≥140 and <90
International Society of Hypertension⁵⁰	
Normal	<130 and <85
High-normal	130–139 or 85–89, or both
Grade 1 hypertension	140–159 or 90–99, or both
Grade 2 hypertension	≥160 or ≥100, or both

Table 1: Classification of office blood pressure by class or grade, mm Hg

	American College of Cardiology–American Heart Association ⁴⁹	European Society of Cardiology–European Society of Hypertension ⁴⁸ and International Society of Hypertension ⁵⁰
Office BPM, mm Hg	<130/80	<140/90
Home BPM, mm Hg	<130/80	<135/85
24 h ambulatory BPM, mm Hg	<125/75	<130/80
Awake ambulatory BPM, mm Hg	<130/80	<135/85
Sleep ambulatory BPM, mm Hg	<110/65	<120/70

BPM=blood pressure monitoring.



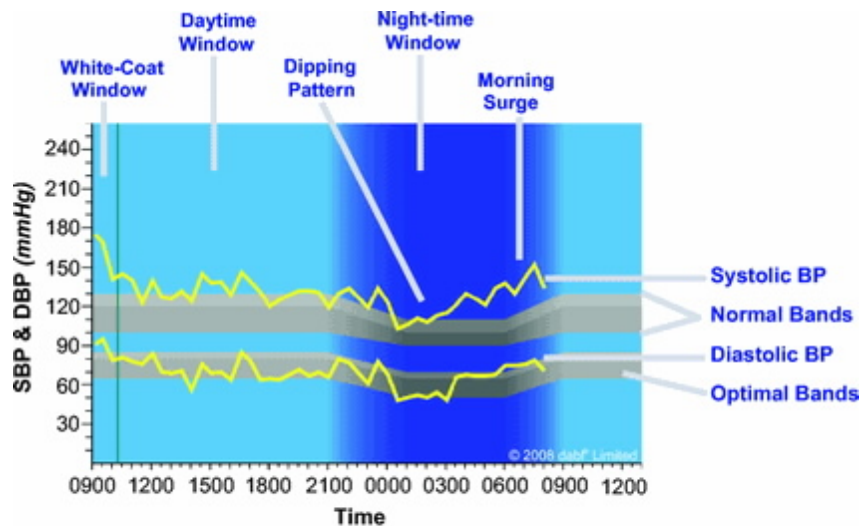
ISH Guideline 2020. Hypertension 2020
Brouwers S, et al. Lancet 2021

Home BP Monitoring

- **Self-measurement of BP at home is of clinical value and its prognostic significance is now demonstrated.**
 - provide more information on the BP lowering effect of treatment at trough, and thus on therapeutic coverage throughout the dose-to-dose time interval
 - improve patient's adherence to treatment regimens
 - there are doubts on technical reliability/environmental conditions of ambulatory BP data
- **Self-measurement of BP at home should be discouraged whenever:**
 - it causes anxiety to the patient
 - it induces self-modification of the treatment regimen
- **Normal values are different for office and home BP (10mmHg lower)**



Indications for ambulatory BP monitoring



The ABPM suggests borderline 24-hour systolic hypertension (133 mmHg daytime, 118 mmHg night-time) and normal 24-hour diastolic blood pressure (71 mmHg daytime, 59 mmHg night-time) with a white-coat effect (175 mmHg/95 mmHg).

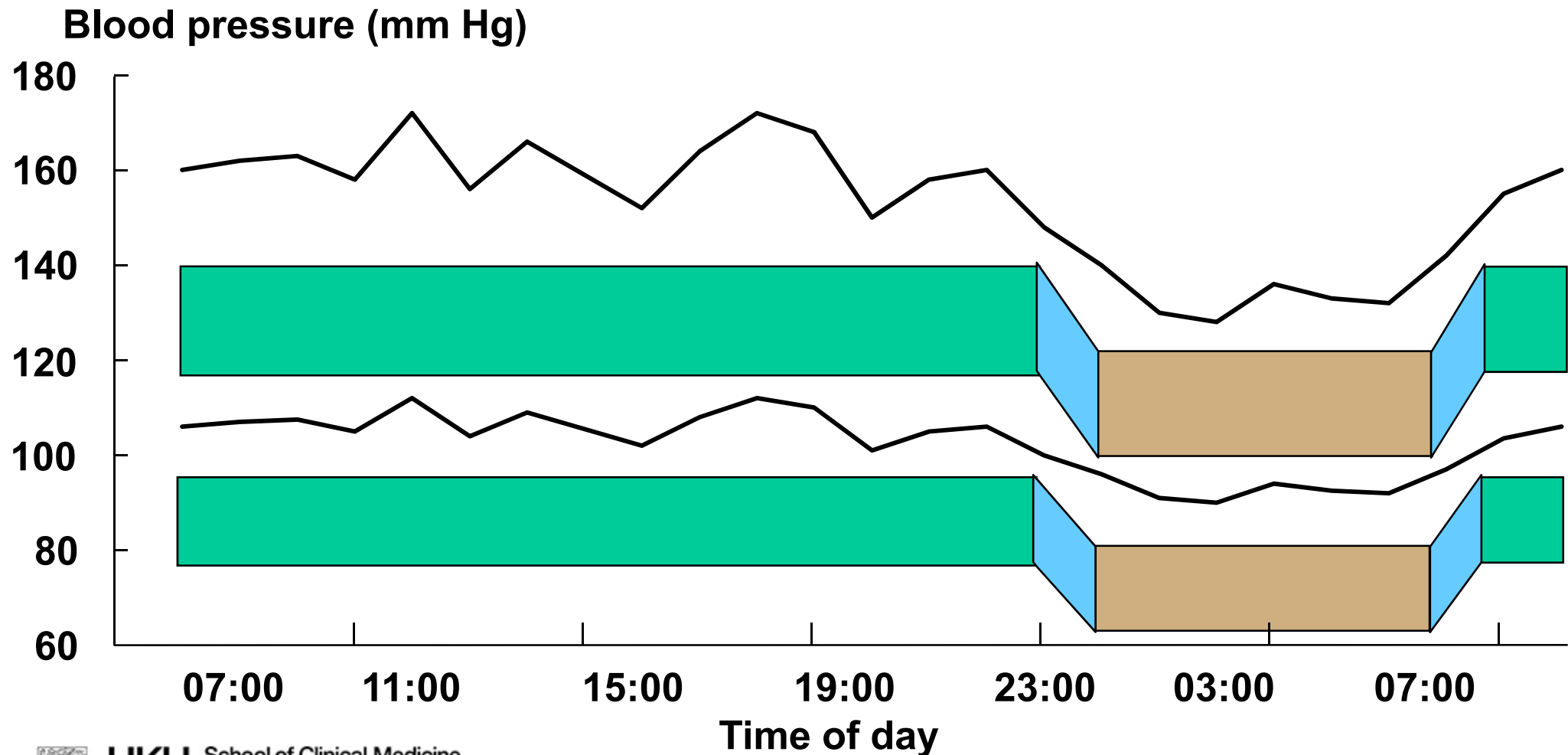
Automatic Interpretation				
Statistical Analysis				
White-Coat Window	SBP	DBP	MAP	HR
Readings	3	3	3	3
First hr max	175	95	124	50
Daytime (09:00-21:00)				
Readings	21	21	21	21
Mean	133	71	90	44
SD	8	7	8	3
Night-time (01:00-06:00)				
Readings	10	10	10	10
Mean	118	59	81	40
SD	9	8	10	2
24-hour				
Readings	43	43	43	43
Mean	129	68	88	42
SD	11	9	10	3
Dip %	11	17	10	8

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- considerable variability of office BP
- marked discrepancy between office BP vs. home BP
- resistance HT
- ? hypotensive episodes esp in elderly and DM
- office BP is elevated in pregnant women and pre-eclampsia is suspected

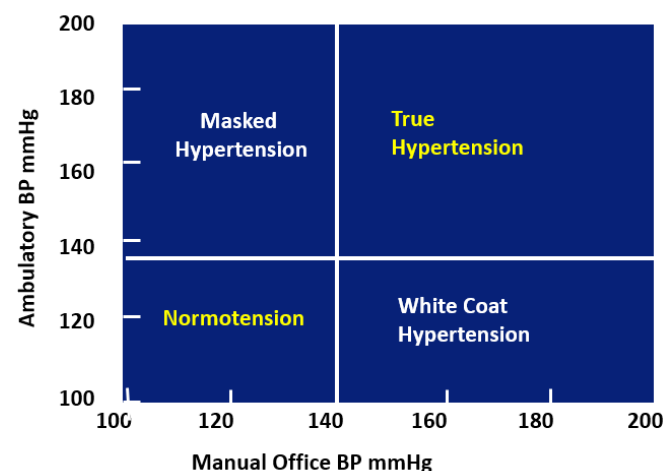


24-h BP profile in a typical hypertensive with an indicative normotensive range



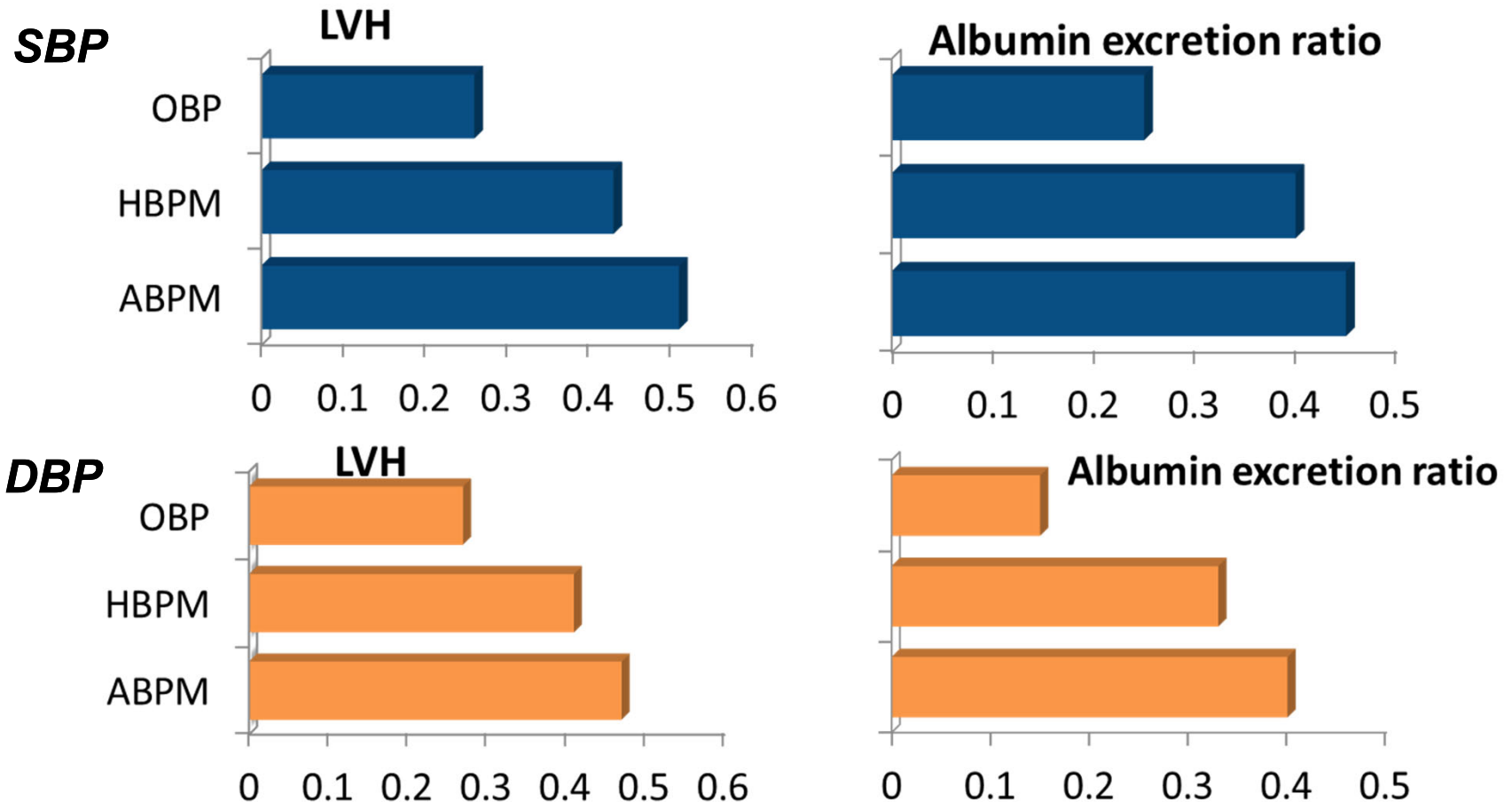
Home BP vs. 24 Hrs Monitoring

	Home Blood Pressure Monitoring	24-Hour Ambulatory Blood Pressure Monitoring
Condition	As for office blood pressure (see above).	Routine working day.
Position	As for office BP (see above).	Avoid strenuous activity. Arm still and relaxed during each measurement.
Device	Validated electronic (oscillometric) upper-arm cuff device (www.stridebp.org , and Section 11: Resources)	
Cuff	Size according to the individual's arm circumference	
Measurement protocol	Before each visit to the health professional: • 3–7-day monitoring in the morning (before drug intake if treated) and the evening. • Two measurements on each occasion after 5 min sitting rest and 1 min between measurements. Long-term follow-up of treated hypertension: • 1–2 measurements per week or month.	• 24-hour monitoring at 15–30 min intervals during daytime and nighttime. • At least 20 valid daytime and 7 nighttime BP readings are required. If less, the test should be repeated.
Interpretation	• Average home blood pressure after excluding readings of the first day ≥ 135 or 85 mm Hg indicates hypertension.	• 24-hour ambulatory blood pressure $\geq 130/80$ mm Hg indicates hypertension (primary criterion). • Daytime (awake) ambulatory blood pressure $\geq 135/85$ mm Hg and nighttime (asleep) $\geq 120/70$ mm Hg indicates hypertension

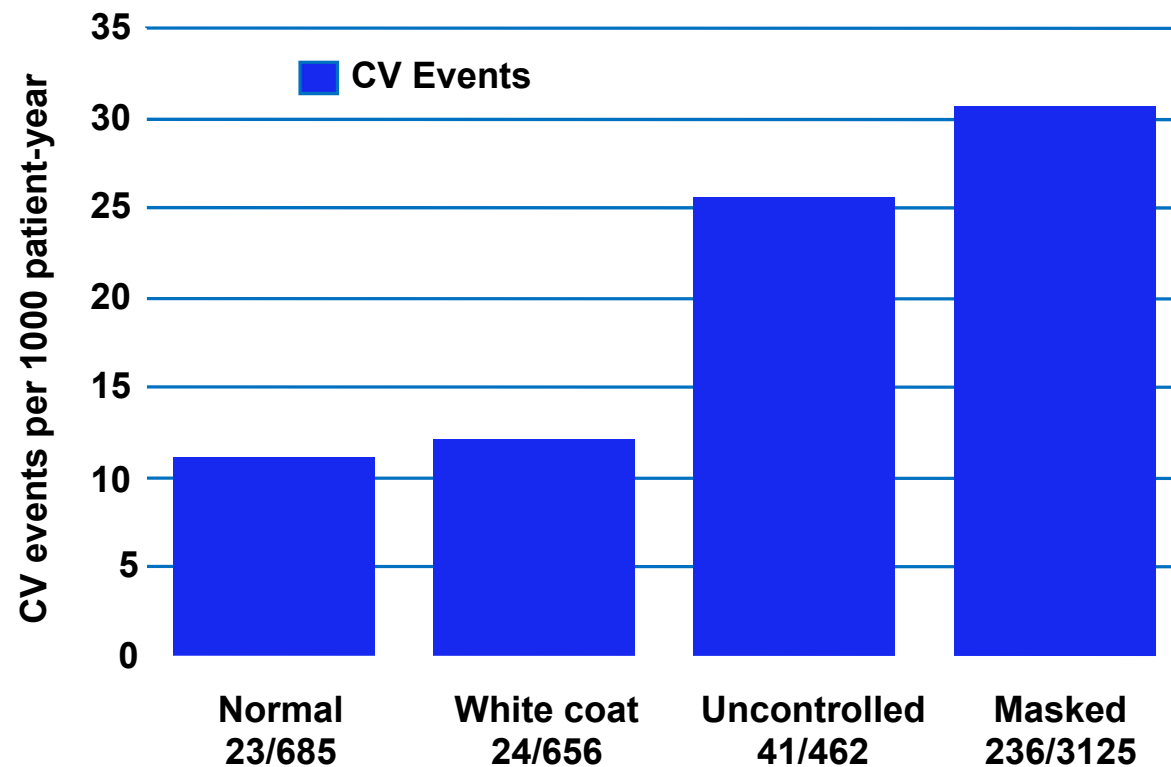


	Home BP	ABP
White coat HT	+	++
Masked HT	+	++
Nocturnal HT	-	++
Liabile BP	+	+
Resistant HT	+	+
Autonomic dysfunction	+	++
Pregnancy HT	++	+

Out-of-office BP measurements are more highly correlated with BP-related risk



The prognosis of *white coat* and *masked* hypertension



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Okhubo et al. J. Am. Coll. Cardiol. 2005;46;508-515

Screening and FU of BP

ABPM	HBPM
Advantages • Can identify white-coat and masked hypertension • Stronger prognostic evidence • Night-time readings • Measurement in real-life settings • Additional prognostic BP phenotypes • Abundant information from a single measurement session, including short-term BP variability	Advantages • Can identify white-coat and masked hypertension • Cheap and widely available • Measurement in a home setting, which may be more relaxed than doctor's office • Patient engagement in BP measurement • Easily repeated and used over longer periods to assess day-to-day BP variability
Disadvantages • Expensive and sometimes limited availability • Can be uncomfortable	Disadvantages • Only static BP is available • Potential for measurement error • No nocturnal readings

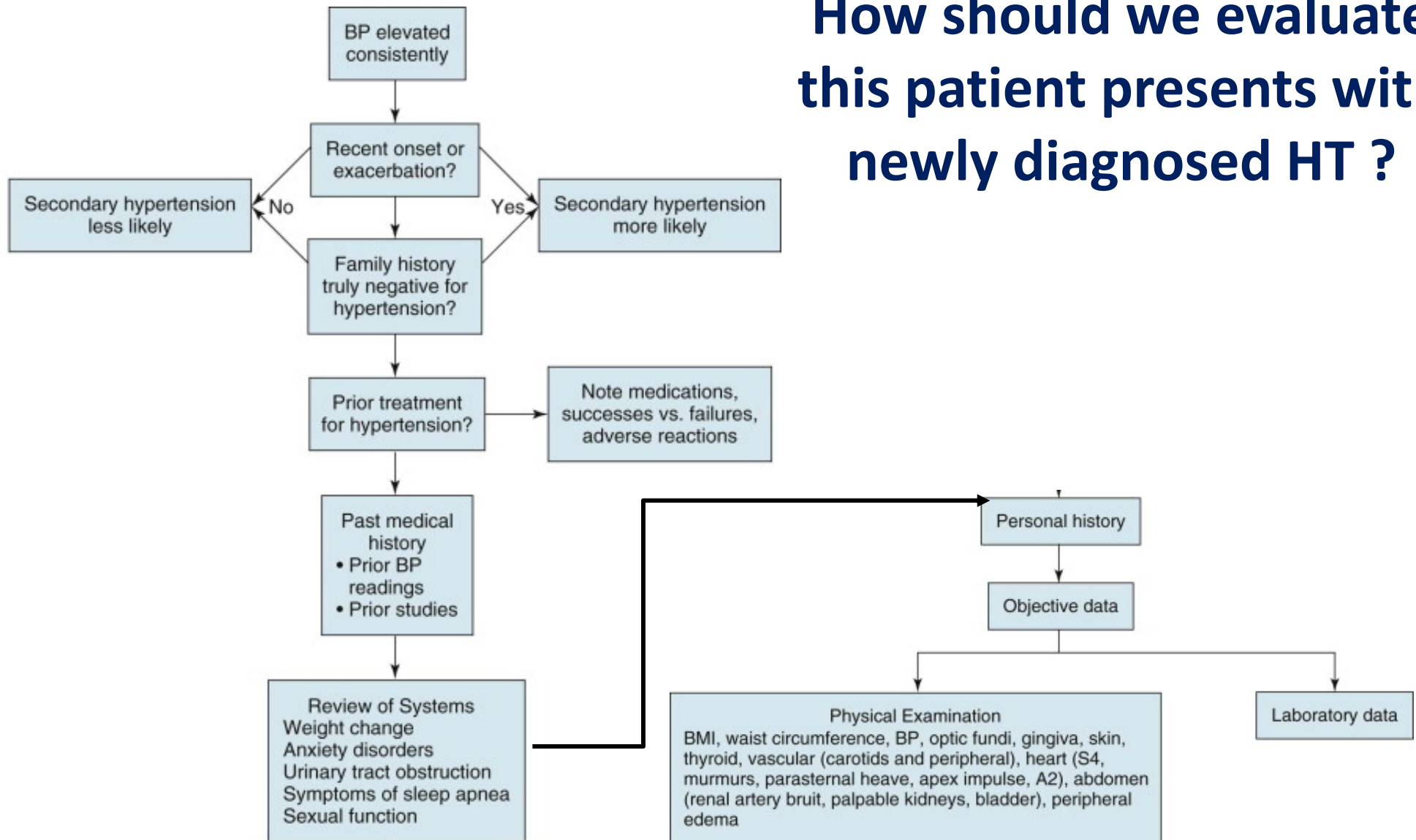


“White-coat” HT

- **Definition:** Office BP is persisted ↑ whereas BP outside clinic is normal, usually diagnosis by ambulatory or home BP monitoring
- **Prevalence:** 15-30% of general population, commonly in elderly and pregnant women
- **Diagnosis:** Identify by home or ambulatory BP measure
- **Risk:** less than sustained HT, probably small risk > than normal people, ? Precursor to HT
- **Implications:**
 - No specific clinical characteristics
 - Must be considered in people with newly diagnosis HT and before treatment, & placed in context of overall risk profile
 - Should reassure patient, employers and insures that the risk is low or absent
 - need close follow-up and monitoring again



How should we evaluate this patient presents with newly diagnosed HT ?



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Cardiovascular Therapeutics: A Companion to Braunwald's Heart Disease

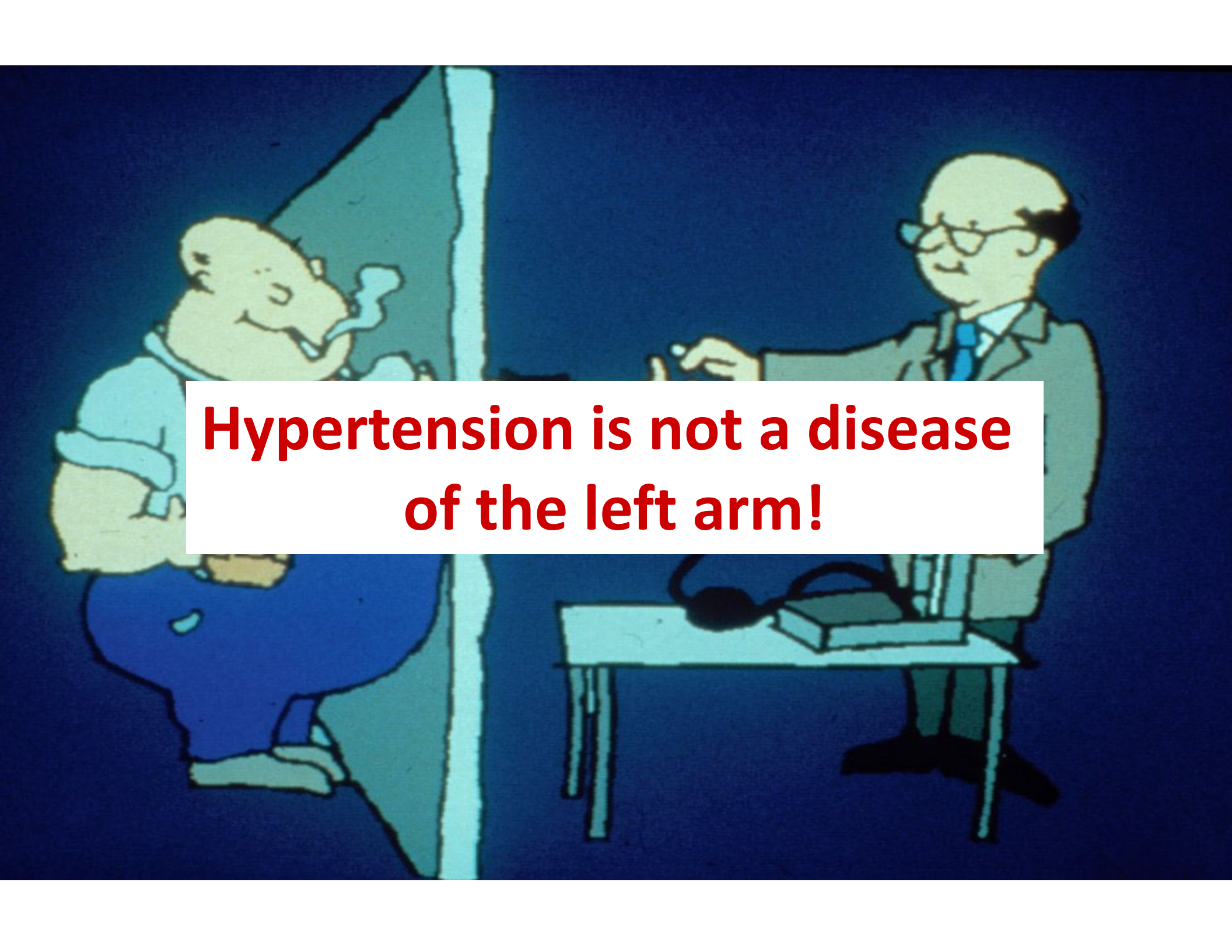
Aims of Clinical Evaluation in Patients with HT

- To confirm a chronic elevation of BP and determine the level
- Uncovering correctable secondary forms of HT
- Establishing a pretreatment baseline,
- Assessing factors that may influence the type of therapy or be changed adversely by therapy,
- Determining if target organ damage is present
- Determining whether other risk factors for the development of atherosclerotic CVS diseases are present



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A cartoon illustration of a doctor and a patient. The doctor, on the right, is a balding man with glasses, wearing a grey suit, white shirt, and blue tie. He is standing behind a white desk, gesturing with his right hand towards the patient. On the desk are a black telephone and a stack of books. The patient, on the left, is a large, balding man with a prominent nose, wearing a light blue shirt and dark blue trousers. He is sitting in a chair, leaning back, and looking towards the doctor. The background is a solid dark blue.

**Hypertension is not a disease
of the left arm!**

Hypertension



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Signs and Symptoms

- **None**
- **Elevated BP itself:** headaches, dizziness, palpitations, easy fatigability, and impotence
- **Hypertensive vascular disease:** epistaxis, hematuria, blurring of vision, episodes of weakness or dizziness, angina pectoris, and dyspnea
- **Underlying disease- secondary HT:** e.g. primary aldosteronism- polyuria, polydipsia, and muscle weakness due to hypo K⁺



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CVS Risk Factors



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CVD Risk Factors

- Hypertension*
- Cigarette smoking
- Obesity* (BMI ≥ 30 kg/m²)
- Physical inactivity
- Dyslipidemia*
- Diabetes mellitus*
- Microalbuminuria or estimated GFR <60 ml/min
- Age (older than 55 for men, 65 for women)
- Family history of premature CVD
(men under age 55 or women under age 65)



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***Components of the metabolic syndrome.**

History-Risks Factors

- **Duration and previous BP levels**
- **Family history of HT: essential HT**
- **Age: secondary HT- 35 or after 55.**
- **Other risk factors:** cigarette smoking, diabetes mellitus, lipid disorders, and a family history of early deaths due to cardiovascular disease.
- **Patient's lifestyle:** diet, physical activity, family status, work, and educational level.



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Physical Examination

- General appearance
- BP and pulses in the two upper extremities (supine and standing)
- Height and weight = BMI, waist circumference
- Fundi examination
- Palpation and auscultation of the all peripheral arteries
- CVS examination



Basic Tests for Initial Evaluation

Test

- Urine examination
- Serum creatinine, urea
- Serum K and Ca
- Serum glucose
- Total cholesterol, HDL, LDL & TG
- Serum uric acid
- Electrocardiogram
- TSH
- Chest x-ray

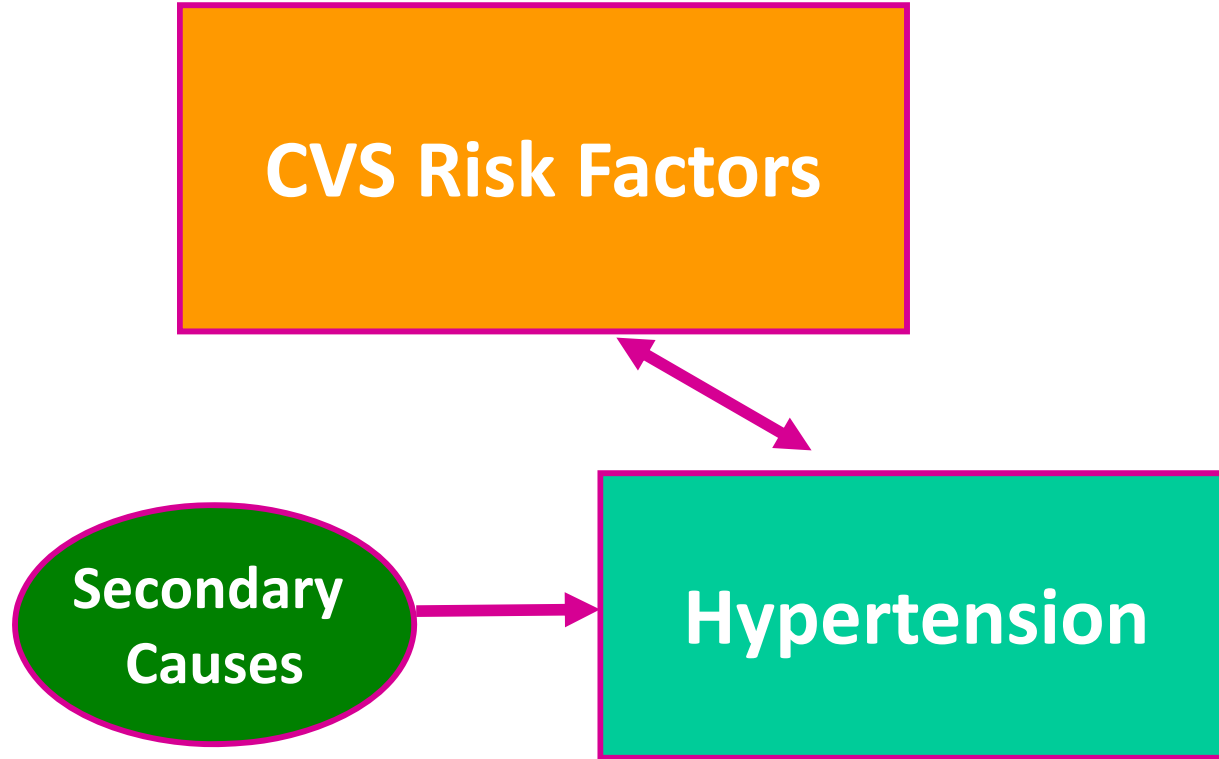
Implications

- To R/O renal diseases
- Baseline kidney function and R/O kidney failure
- ? Hyperaldosteronism and hyperparathyroidism
- R/O diabetes
- Look for associated CVS risk factors
- Baseline and ? gout
- LVH, MI, cardiac failure, heart block
- To R/O thyroid disorders
- ? Cardiomegaly and cardiac failure



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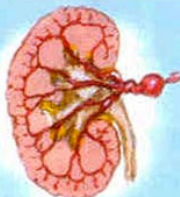
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Etiology of Secondary HT

Common

Uncommon

Renal		Parenchymal renal disease Glomerulonephritis Chronic pyelonephritis Diabetic nephropathy Polycystic kidneys Connective-tissue diseases Gouty nephritis Renal amyloidosis Radiation nephritis Tuberculosis Hypernephroma Other renal tumors Perinephritis Renal hematoma
		Occlusive vascular disease Atherosclerotic obstruction of renal artery or branches Fibromuscular hyperplasia of renal artery Thrombotic obstruction of renal artery or branches Embolic obstruction of renal artery or branches Other diseases of renal artery (hypoplasia, inflammation, pseudoxanthoma elasticum, etc.)
Adrenal		Pheochromocytoma Primary hyperaldosteronism Cushing's syndrome Adrenogenital syndrome
Neurogenic		Brain tumor Epilepsy Increased intracranial pressure Encephalitis Ganglioneuroma Tabes dorsalis Diencephalic syndrome Lead encephalitis Bulbar poliomyelitis Neuroblastoma Transection of cord Polyneuritis Porphyrria

Coarctation of aorta			
Toxemia of pregnancy		Preeclampsia	Eclampsia
Other causes		Hypercalcemia (hyperparathyroidism) Oral contraceptives Licorice ingestion	Amphetamines and other vasopressor drugs Monamine oxidase inhibitors
Caused mainly by increased stroke volume of left ventricle		Complete heart block Aortic regurgitation Patent ductus arteriosus Thyrotoxicosis	Arteriovenous fistula Severe anemia Beriberi Paget's disease of bone
Caused mainly by decreased distensibility of aorta		Arteriosclerosis of aorta Coarctation of aorta (see above)	

Diagnosis	General Population	Special Clinic
Essential HT	92-94 %	65-85 %
Renal HT	1-3 %	4-16 %
Endocrine HT	1 %	1-12 %
Other	0.2 %	1%



Indications of the need to search for secondary HT

Characteristics

Younger patients (< 40 years) with grade 2 hypertension or onset of any grade of hypertension in childhood.

Acute worsening hypertension in patients with previously documented chronically stable normotension.

Resistant hypertension.

Severe (grade 3) hypertension or a hypertension emergency.

Presence of extensive HMOD.

Clinical or biochemical features suggestive of endocrine causes of hypertension or CKD.

Clinical features suggestive of obstructive sleep apnoea.

Symptoms suggestive of pheochromocytoma or family history of pheochromocytoma.



History-Secondary HT

- Family history of renal disease (**polycystic kidney**)
- Renal disease, urinary tract infection, haematuria, analgesic abuse (**parenchymal renal disease**)
- Drug/substance intake:
- Snoring, daytime somnolence, obesity (**OSA**)
- Episodes of sweating, headache, anxiety, palpitation (**phaeochromocytoma**)
- Episodes of muscle weakness and tetany (**aldosteronism**)
- Kidney stone, depression, lethargy, weakness (**Hyperparathyroidism**)
- Fatigue, wt gain, hair loss, diastolic HT, weakness (**hypothyroidism**)
- Heat intolerance, wt loss, tremor, palpitation (**hyperthyroidism**)



P/E: Secondary HT

- Features of Cushing syndrome
- Skin stigmata of neurofibromatosis (**phaeochromocytoma**)
- Palpation of enlarged kidneys (**polycystic kidney**)
- Auscultation of abdominal murmurs (**renovascular hypertension**)
- Auscultation of precordial or chest murmurs (**aortic coarctation or aortic disease**)
- Diminished and delayed femoral pulses and reduced femoral BP (**aortic coarctation, aortic disease**)



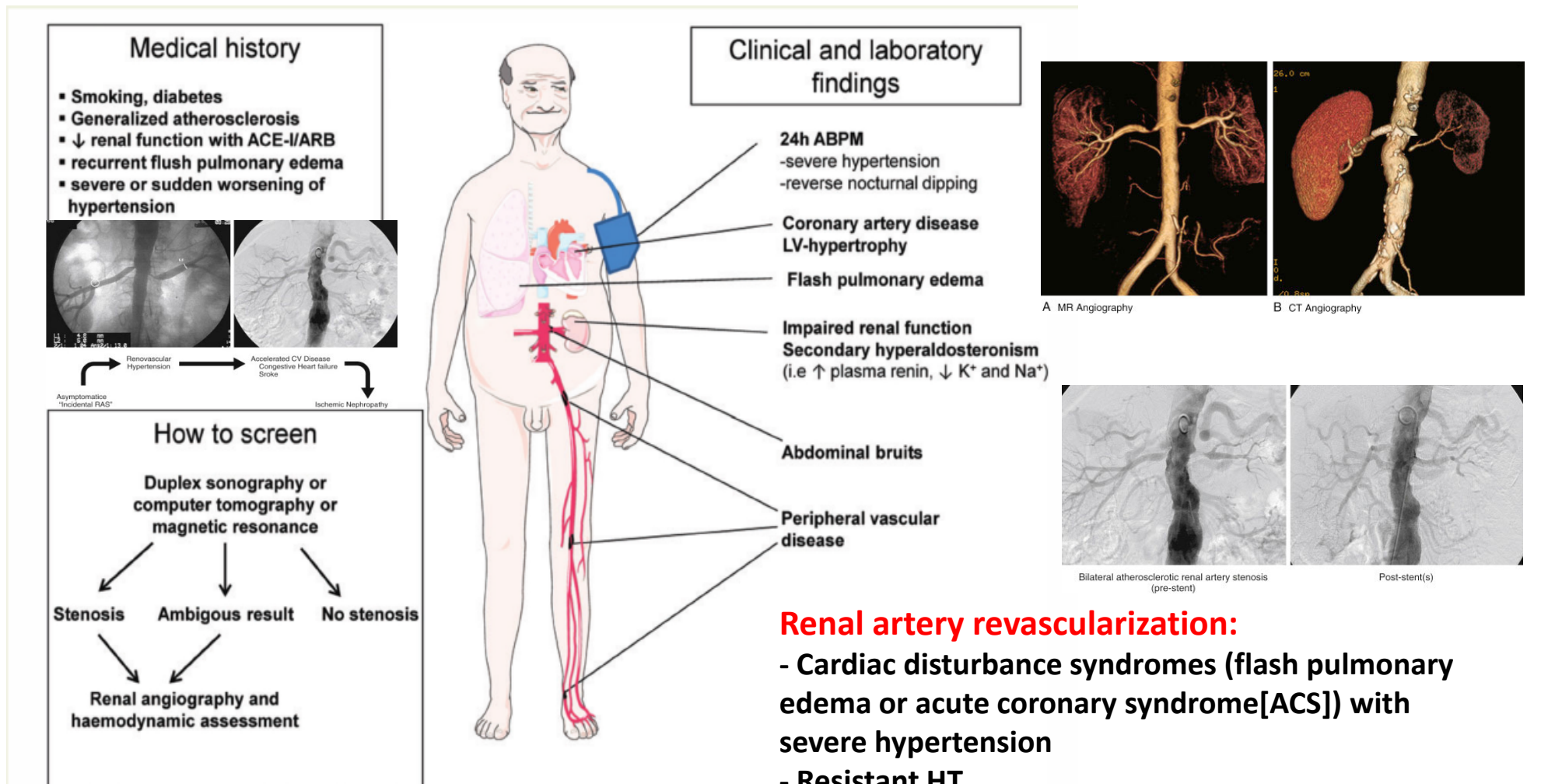
Screening Tests for Secondary HT

	Possible diagnosis	Diagnostic tests
Different blood pressure ($\geq 20/10$ mm Hg) between upper and lower extremities, right and left arms, or both; delayed femoral pulsations, interscapular ejection murmur, rib notching on chest radiograph	Coarctation of the aorta	Echocardiography, chest radiograph, MRI
Peripheral oedema, pallor, loss of muscle mass	Renal parenchymal disease	Creatinine, ultrasound of the kidney
Abdominal bruits, peripheral vascular disease	Renal artery stenosis	Duplex, CT, MRI, angiography
Fatigue, constipation, polyuria, polydipsia, muscle weakness	Primary aldosteronism	Aldosterone–renin ratio
Weight gain, impotence, fatigue, psychological changes, polydipsia and polyuria, obesity, hirsutism, skin atrophy, striae rubrae, muscle weakness, osteopenia	Cushing's syndrome	24 h urinary cortisol, dexamethasone testing
Headache, palpitations, flushing, anxiety, paroxysmal hypertension, pounding, headache, perspiration, palpitations, pallor	Pheochromocytoma	Plasma or 24 h urinary metanephrines, 24 h urinary catecholamine
Palpitations, weight loss, anxiety, heat intolerance, tachycardia, atrial fibrillation, accentuated heart sounds, exophthalmos	Hyperthyroidism (thyroid disease)	Thyroid stimulating hormone, tri-iodothyronine, thyroxine
Weight gain, fatigue, obstipation, bradycardia, muscle weakness, myxoedema	Hypothyroidism (thyroid disease)	Thyroid stimulating hormone, tri-iodothyronine, thyroxine
Snoring, daytime sleepiness, morning headache, irritability, increase in neck circumference, obesity, peripheral oedema	Obstructive sleep apnoea	Screening questionnaire, polysomnography

Modified from Sudano and colleagues.⁶⁴

Table 3: Signs and symptoms suggesting specific causes of secondary hypertension

Renovascular Hypertension



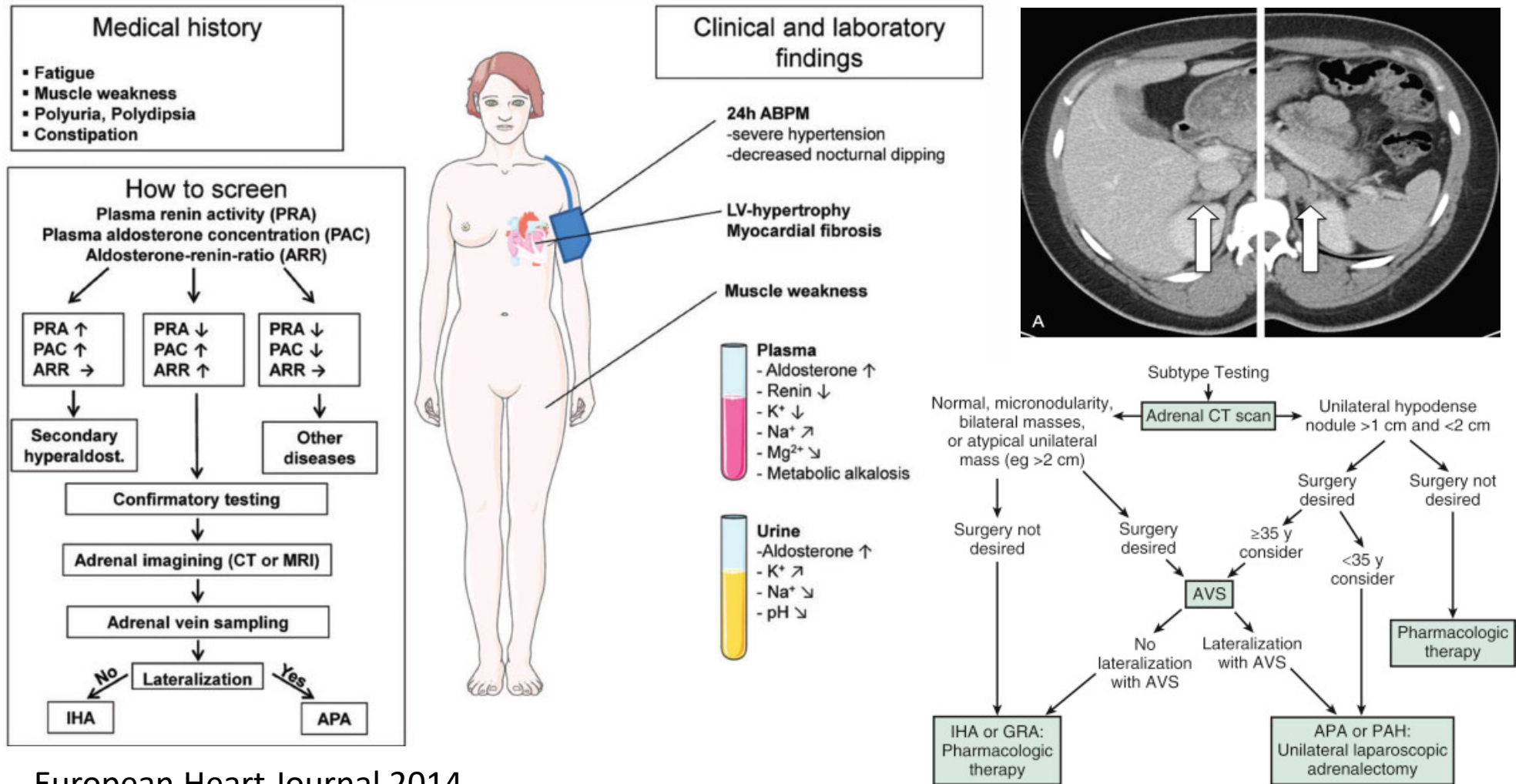
European Heart Journal 2014



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Mineralocorticoid Hypertension



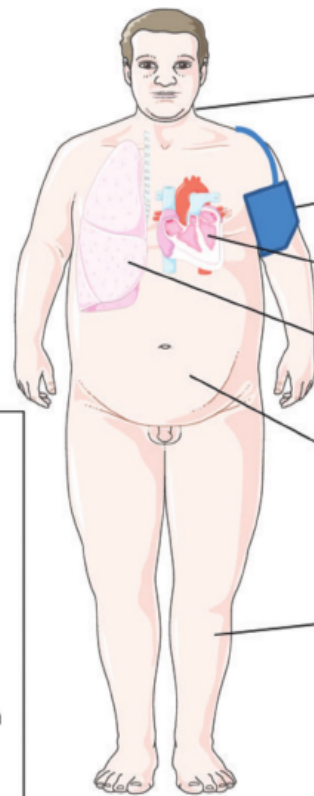
European Heart Journal 2014

Obstructive Sleep Apnea

Medical history

- Snoring
- Exaggerated daytime sleepiness
- Morning headaches
- Lack of concentration
- Irritability
- Increased frequency of motor vehicle accidents

Clinical findings



Narrow upper airway
-i.e. large neck (m > 42; w > 39 cm)
-macroglossia

24h ABPM
-non-dipper
-tachycardia and/or bradycardia

Left ventricular hypertrophy
Cor pulmonale

Pulmonary hypertension

Obesity

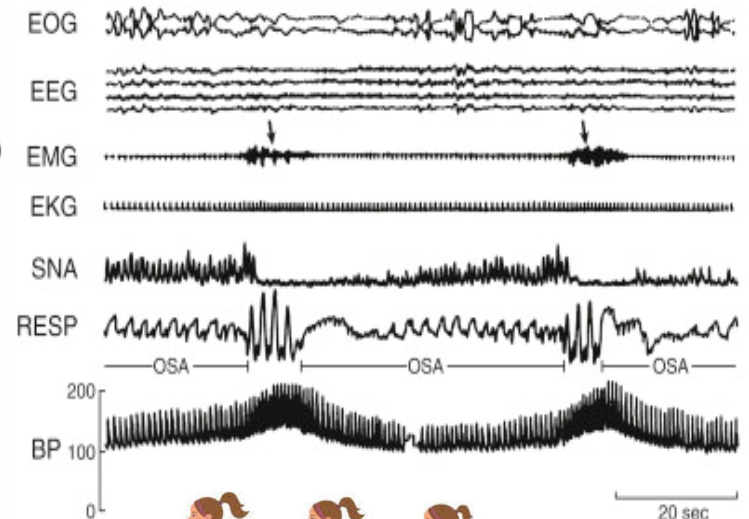
Peripheral edema

How to screen

Screening questionnaire (i.e. Epworth)

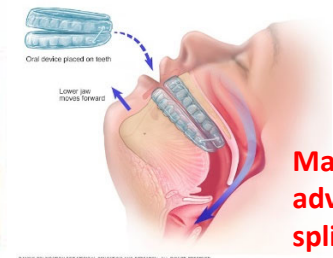
Polysomnography

AHI 5-15/h → mild
AHI 15-30/h → moderate
AHI > 30/h → severe



Weight reduction

CPAP



Mandibular advancement splints

European Heart Journal 2014



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Hypertension: A Companion to Braunwald's Heart Disease

Medications and Hypertension

Oestrogen

- Oral contraceptives

Herbal

- Ephedra (Ma huang)
- Ginseng

Illicit

- Amphetamines
- Cocaine

Non-steroidal anti-inflammatory

- COX-2 inhibitors
- Ibuprofen
- Naproxen

Psychiatric

- Buspirone
- Carbamazepine
- Clozapine
- Fluoxetine
- Lithium
- Tricyclic antidepressants

Steroid

- Methylprednisolone
- Prednisone

Sympathomimetic

- Decongestants
- Diet pills



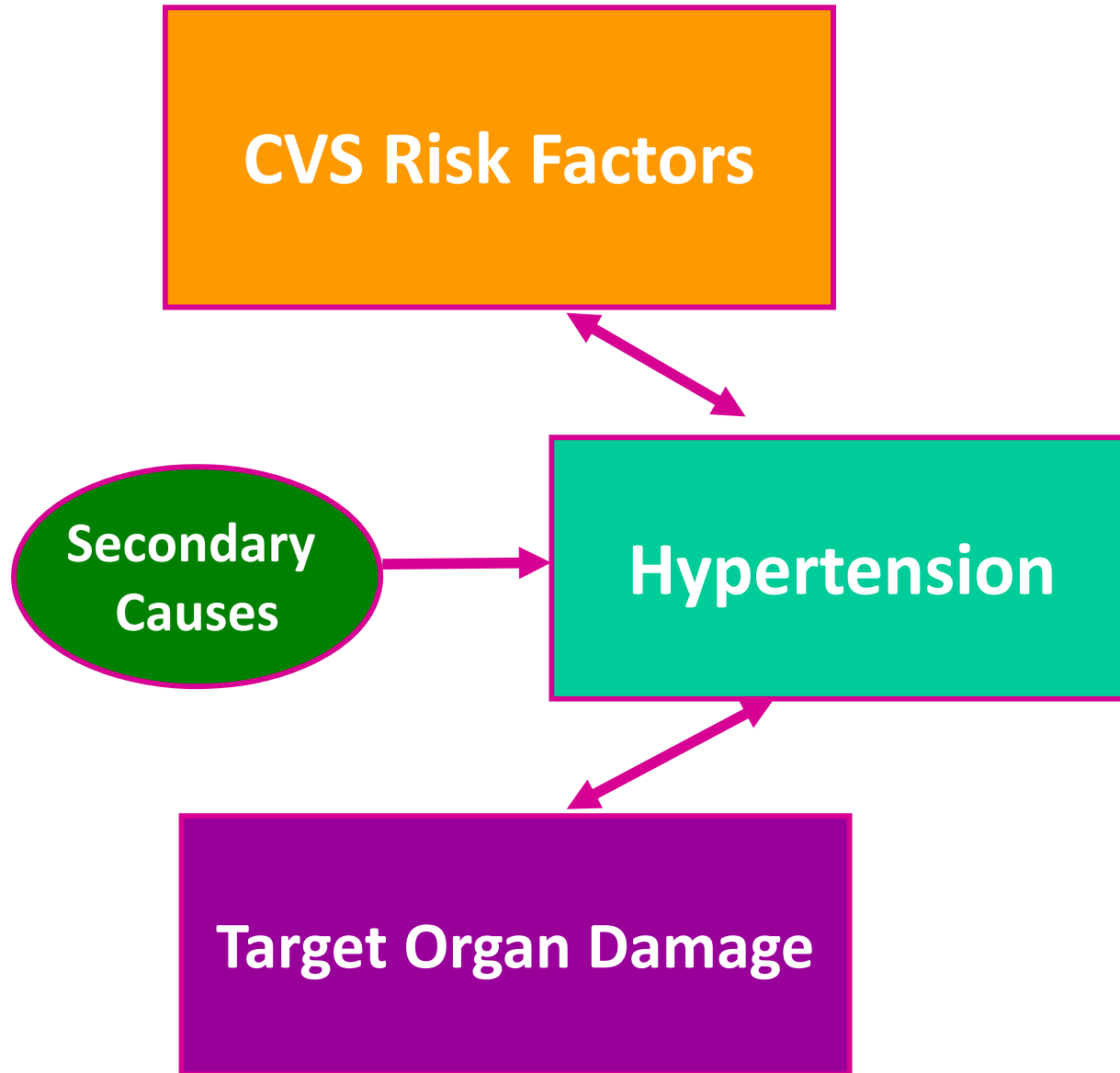
Identifiable Causes of HT

- **A- Accuracy of diagnosis, Apnea-Obstructive sleep apnea**
- **B- Bruit (renal artery stenosis) or Bad kidney (Renal failure, polycystic kidney)**
- **C- Catecholamine (phaeochromocytoma), Cushing and Coarctation of aorta**
- **D- Diet (salt/alcohol/obesity) and Drugs (immunosuppressive agents, NSAID, COX-2 inhibitors, estrogens / oral contraceptive, weight-loss agents, stimulants, mineralocorticoids, antiparkinsonian, monoamine oxidase inhibitors, anabolic steroids, sympathomimetics)**
- **E- Erythropoietin and Endocrine (primary and secondary hyperaldosteronism, hyper- or hypothyroidism, parathyroid disease)**



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History-TOD/CVD

- **Brain and eyes:** headache, vertigo, impaired vision, transient ischaemic attacks, sensory or motor deficit
- **Heart:** palpitation, chest pain, shortness of breath, swollen ankles
- **Kidney:** thirst, polyuria, nocturia, haematuria
- **Peripheral arteries:** cold extremities, intermittent claudication



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P/E: Target Organ Damage

- **Brain:** murmurs over neck arteries, motor or sensory defects
- **Retina:** fundoscopic abnormalities
- **Heart:** location and characteristics of apical impulse, abnormal cardiac rhythms, ventricular gallop, pulmonary rales, peripheral oedema
- **Peripheral arteries:** absence, reduction, or asymmetry of pulses, cold extremities, ischaemic skin lesions
- **Carotid arteries:** systolic murmurs



Identifying target organ damage

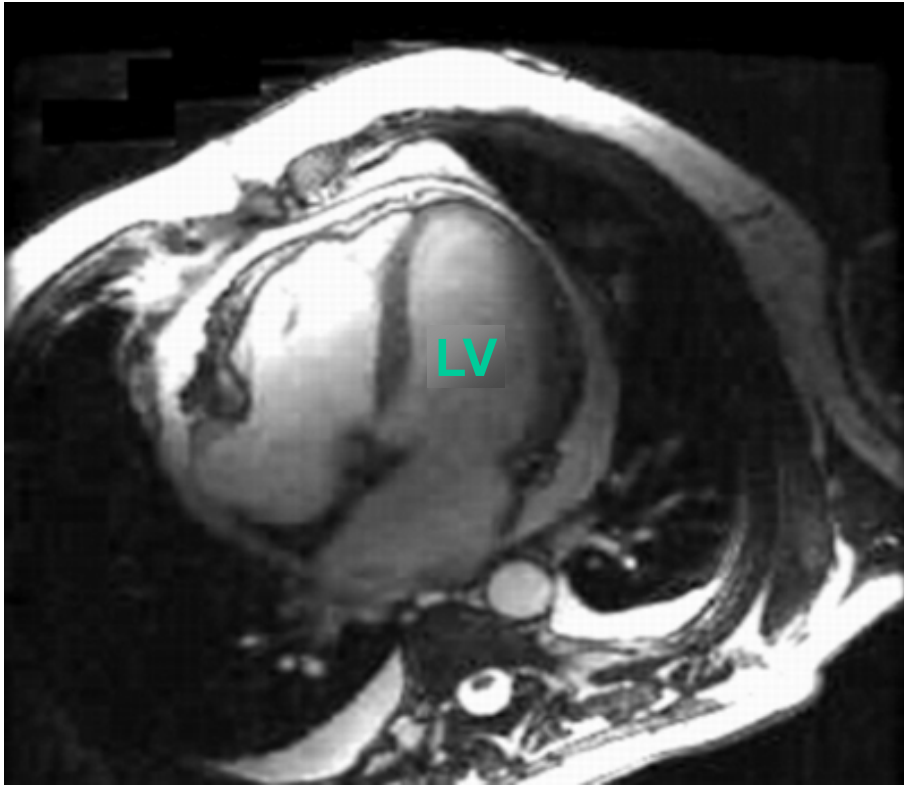
Organ	Condition	Investigations
Heart	Left ventricular hypertrophy, diastolic dysfunction	ECG, Echo, CXR
Kidney	Renal impairment, proteinuria	Serum creatinine level and dip stix
Large arteries	Atherosclerosis plaque (aorta, carotid, iliac, femoral)	Ultrasound
Eye fundi	Narrowing of retinal artery	Fundoscopy



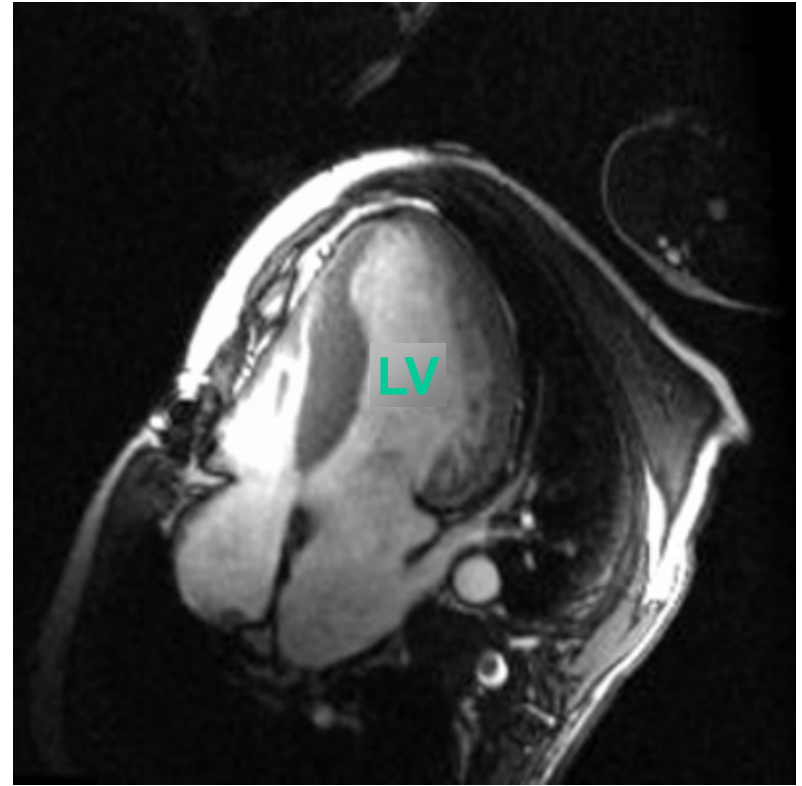
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Diagnosis of LVH Imaging: ECG/Echo/MRI



Normal heart



Heart with LVH



Risk Factors for an Adverse Prognosis in Hypertension

- Black race
- Youth
- Male sex

Concomitant CVS Risk Factors

- Obesity
- Excess alcohol intake

Evidence of end organ damage

Cardiac

- Cardiac enlargement

Target organs damage

- Papilledema

Renal

- impaired renal function

Nervous system:

- Cerebrovascular accident

CVS Risk Factors

**Secondary
Causes**

Hypertension

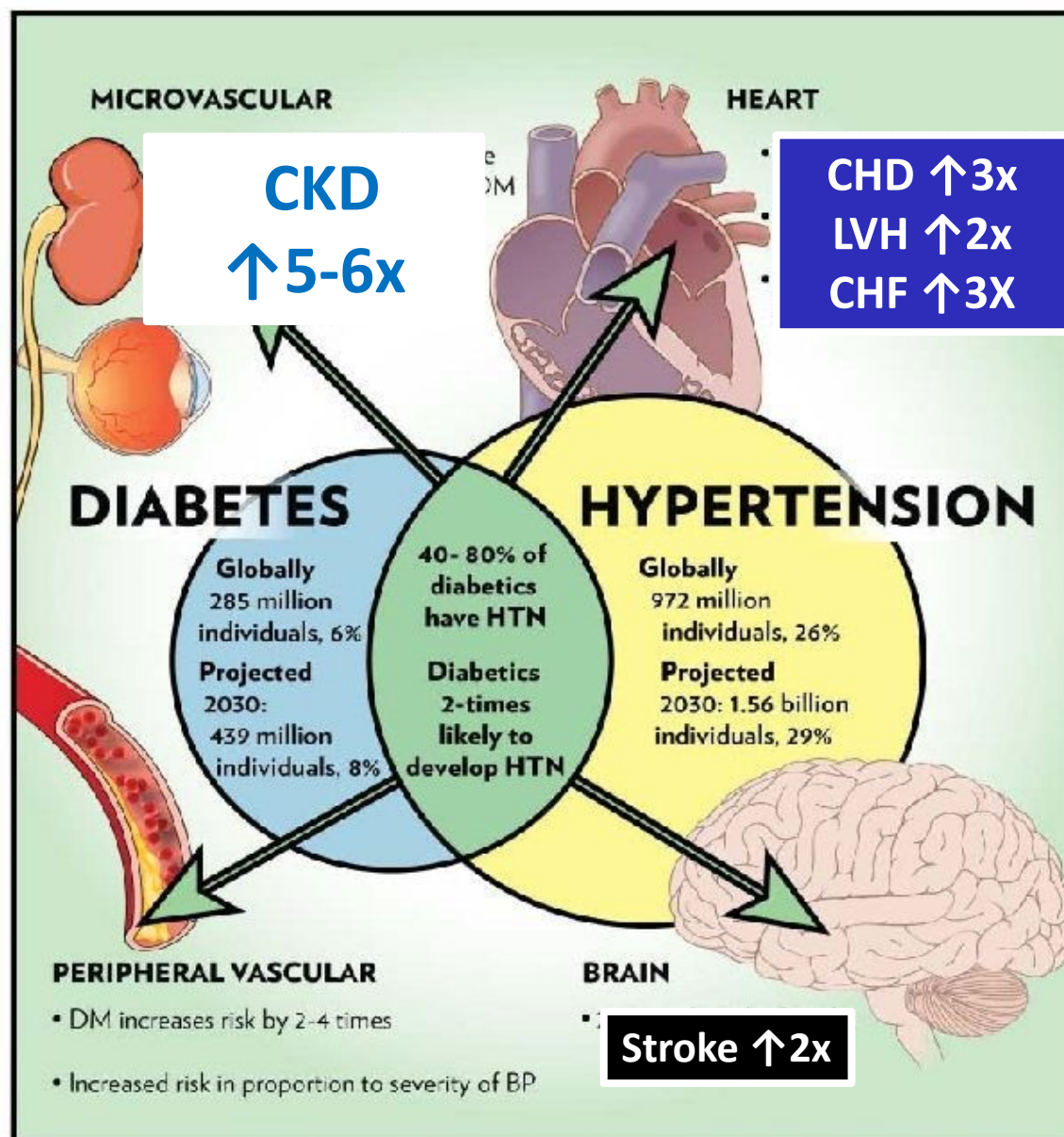
Target Organ Damage



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Diabetes and Hypertension: Bad Companions



Goal of Therapy

- To achieve the maximum reduction in the total risk of CVD- need for treatment of other risk factor + BP control
- To achieve optimal or normal BP in young middle aged, or diabetic patients (below 140/85-80 mmHg) and high normal BP in elderly patients (below 150-140/90 mmHg)
- Successful management= good communication and relationship between patient and physician as treatment for hypertension is for life



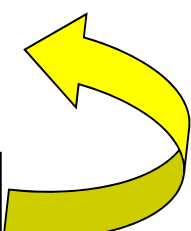
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Implement Lifestyle Modification

**Is BP above goal?
($<150/90$ or $<140/80$ for young (<65), DM and renal diseases)**

No



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Lifestyle Measures

Normotensive individuals:

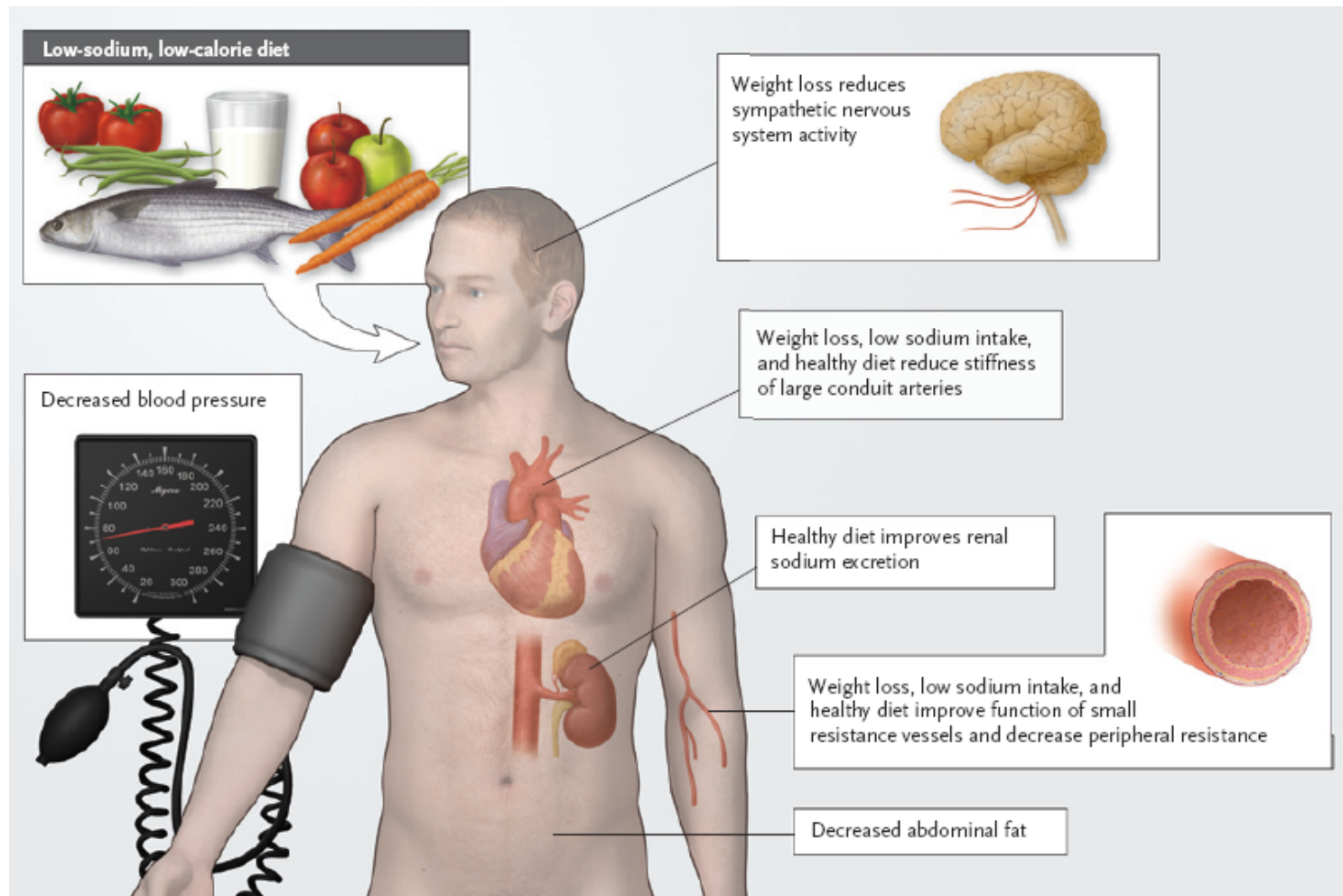
- to prevent the onset of hypertension
- to reduce blood pressure → ↓ CVS risk

Hypertensive patients:

- can serve as initial therapy before medication
- an adjunct to medication
- can facilitate medication step-down or even withdrawal in certain individuals
- ↓ CVS risk



Diet control for HT



Impact of interventions on BP

Interventions	Mean SBP reduction (mmHg)
A) Lifestyle	
• Weight reduction if BMI >25-10 kg wt loss	5-20
• Halving salt intake (10 to 5 g/d)	2-8
• Regular exercise (brisk walking 30min/d)	4-9
• Diet rich in fruit, vegetables, and low fat dietary product with low total fat (DASH diet)	8-14
• Moderation of alcohol consumption <2 drinks per day	2-4
B) Drug Treatment	
• Thiazide	15
• Beta-blocker	10
• Calcium channel blockers	10



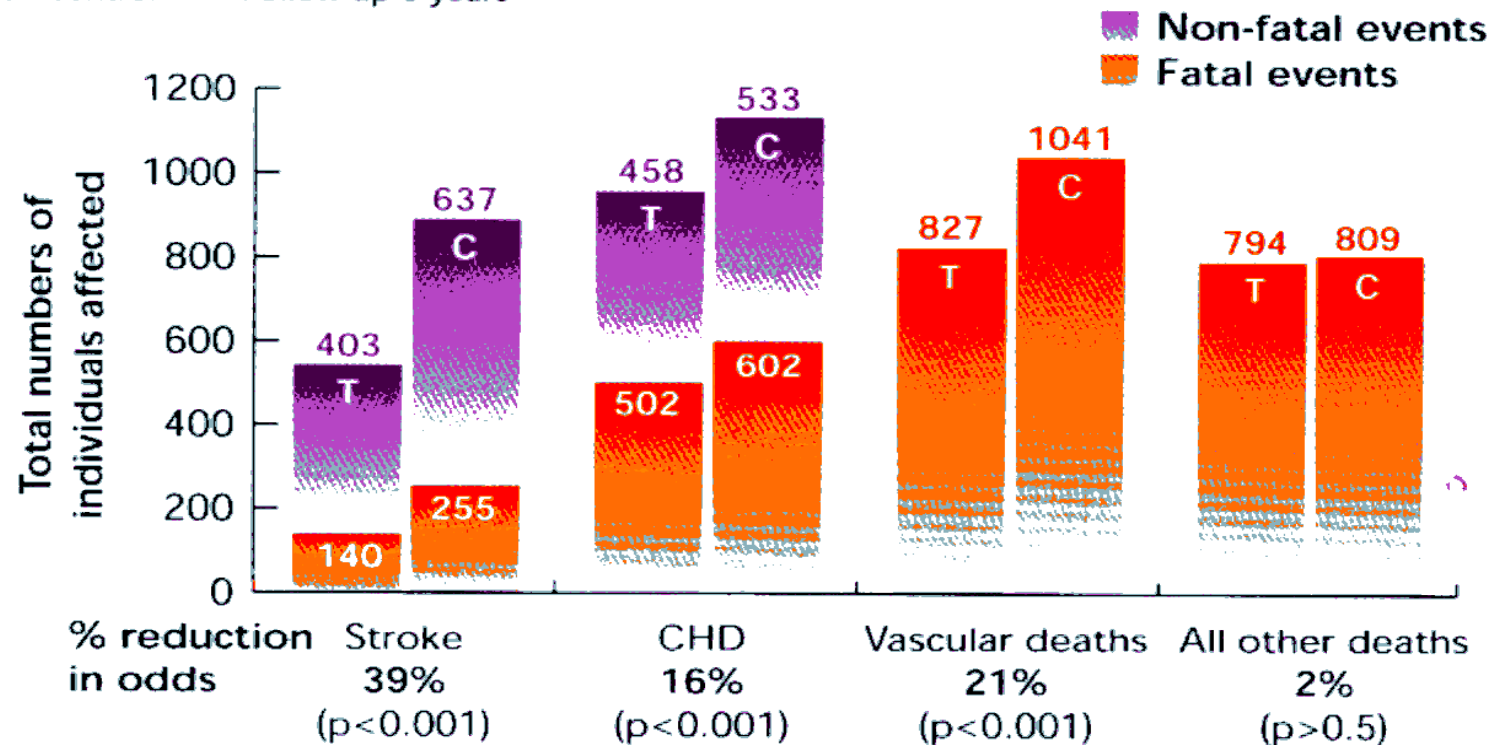
Lifestyle Measures

	American College of Cardiology– American Heart Association 2017 ⁴⁹	European Society of Cardiology– European Society of Hypertension 2018 ⁴⁸	National Institute for Health and Care Excellence 2019 ⁶⁷	International Society of Hypertension 2020 ⁵⁰
Sodium consumption	<1500 mg/day (ideal)	Limit to up to 2000 mg/day	Encourage a reduction in sodium intake	Avoid foods with high salt content
Diet	DASH	High in fruits and vegetables, low-fat dairy, fish, whole grains; low in red meat and saturated fats	Encourage a healthy diet	DASH diet; high in fruits and vegetables, polyunsaturated fats, and dairy; low in foods high in sugar, saturated fats, and trans fats
Alcohol consumption	≤2 standard drinks* per day for men; ≤1 standard drinks* per day for women	<14 units/week for men†; <8 units/week for women†	Encourage a reduction in intake if excessive	≤2 standard drinks‡ per day for men; ≤1.5 standard drinks‡ per day for women
Physical activity	Aerobic exercise: 90–150 mins/week; dynamic resistance training: 90–150 mins/week	Aerobic exercise for ≥30 mins/day ≥5 days/week	Encourage regular exercise	Moderate aerobic activity for 30 mins/day ≥5 days/week or high-intensity interval training; resistance or strength exercises 2–3 days/week
Weight reduction	Target ideal bodyweight	Avoid BMI >30 kg/m ² or waist circumference >102 cm in men or >88 cm in women	NA	Ethnic-specific BMI and waist circumference cutoffs to avoid obesity

DASH=dietary approach to stop hypertension. NA=not applicable. BMI=body-mass index. *One standard drink contains roughly 14 g of pure alcohol, which is typically found in 12 oz of regular beer (usually about 5% alcohol), 5 oz of wine (usually about 12% alcohol), and 1.5 oz of distilled spirits (usually about 40% alcohol). †1 unit is equal to 125 mL of wine or 250 mL of beer. ‡One standard drink is equivalent to 10 g of alcohol.

Effects of Anti-HT Treatment in Randomized Controlled Trial

T = treatment 52,348 patients: difference in SBP 10-12 mm Hg, difference in DBP 5-6 mm Hg
C= control Follow-up 5 years



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Collins et al. Lancet 1990

CVD Risk Factors

Hypertension disease staging	Other risk factors, HMOD, or disease	BP (mmHg) grading			
		High-normal SBP 130–139 DBP 85–89	Grade 1 SBP 140–159 DBP 90–99	Grade 2 SBP 160–179 DBP 100–109	Grade 3 SBP ≥ 180 DBP ≥ 110
Stage 1 (uncomplicated)	No other risk factors	Low risk	Low risk	Moderate risk	High risk
	1 or 2 risk factors	Low risk	Moderate risk	Moderate – high risk	High risk
	≥ 3 risk factors	Low – moderate risk	Moderate – high risk	High risk	High risk
Stage 2 (asymptomatic disease)	HMOD, CKD grade 3, or diabetes mellitus without organ damage	Moderate – high risk	High risk	High risk	High – very high risk
Stage 3 (symptomatic disease)	Symptomatic CVD, CKD grade ≥ 4, or diabetes mellitus with organ damage	Very high risk	Very high risk	Very high risk	Very high risk



How benefits of treating hypertension?

Problems and Rx	Events prevented	NNT for 5 yrs to prevent 1 event
Rx for severe diastolic HT (115-129 mmHg)	Death, stroke, myocardial infarction	3
Rx for mild-mod diastolic HT (90-109 mmHg)	Death, stroke, myocardial infarction	141
Rx for uncomplicated HT, no risk assessment (age 70-84)	Stroke	34
Rx for uncomplicated HT, no risk assessment (age >60)	Stroke	43
Rx for uncomplicated HT, no risk assessment (age 36-64)	Stroke	850
Rx if 10 yrs CHD risk is 30% (CVS risk 40%)	Cardiovascular event	20
Rx if 10 yrs CHD risk is 30% (CVS risk 40%)	Cardiovascular event	40

Treatment Goal

Blood pressure targets

Clinic blood pressure

- People aged under 80 years: lower than 140/90 mmHg
- People aged over 80 years: lower than 150/90 mmHg

Daytime average ABPM or average HBPM blood pressure during the person's usual waking hours

- People aged under 80 years: lower than 135/85 mmHg
- People aged over 80 years: lower than 145/85 mmHg

Interventions to support adherence to treatment

- Only use interventions to overcome practical problems associated with non-adherence if a specific need is identified²⁵.
- Target the intervention to the need. Interventions might include:
 - suggesting that people record their medicine-taking
 - encouraging people to monitor their condition
 - simplifying the dosing regimen
 - using alternative packaging for the medicine
 - using a multi-compartment medicines system²⁵.



Risk and Benefit of Aggressive BP Lowering

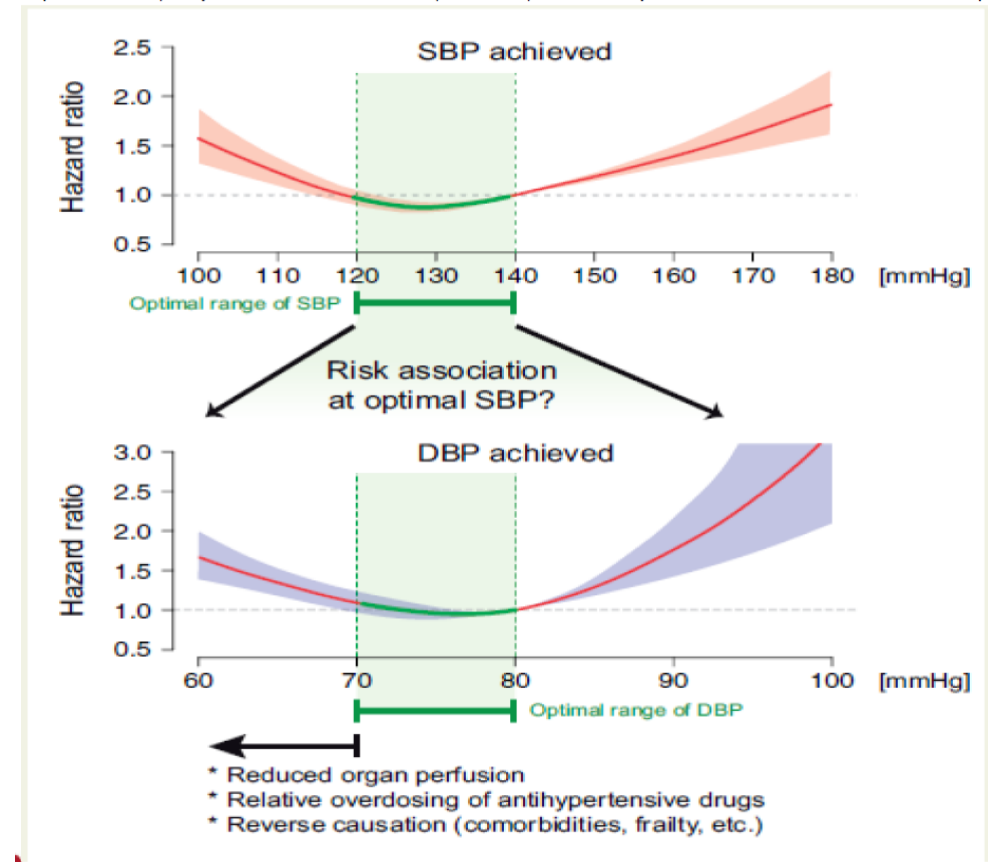
Risk reduction (%) by 10 mmHg SBP reduction at initial SBP of 130-139 mmHg (RCTs)

CHD +	-12
Stroke *	-27
HF *	-25
Major CV events *	-13
All-cause mortality *	-11



Williams B, European Society of Cardiology 2022

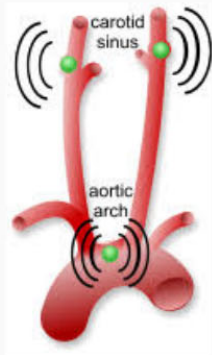
(CV death, myocardial infarction, stroke, and hospitalization for heart failure)



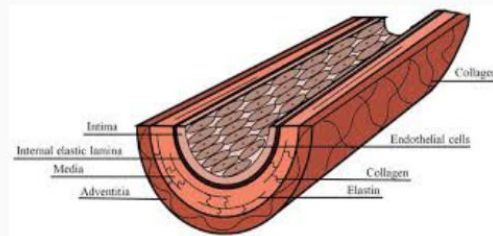
Blood Pressure Target

Why don't we just reduce blood pressure to normal physiological levels?

Because most of our patients don't have normal physiology....



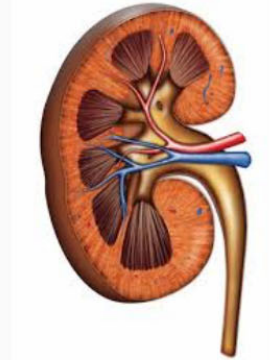
Impaired Baroreceptor



Stiff Arteries



Atheromatous Arteries



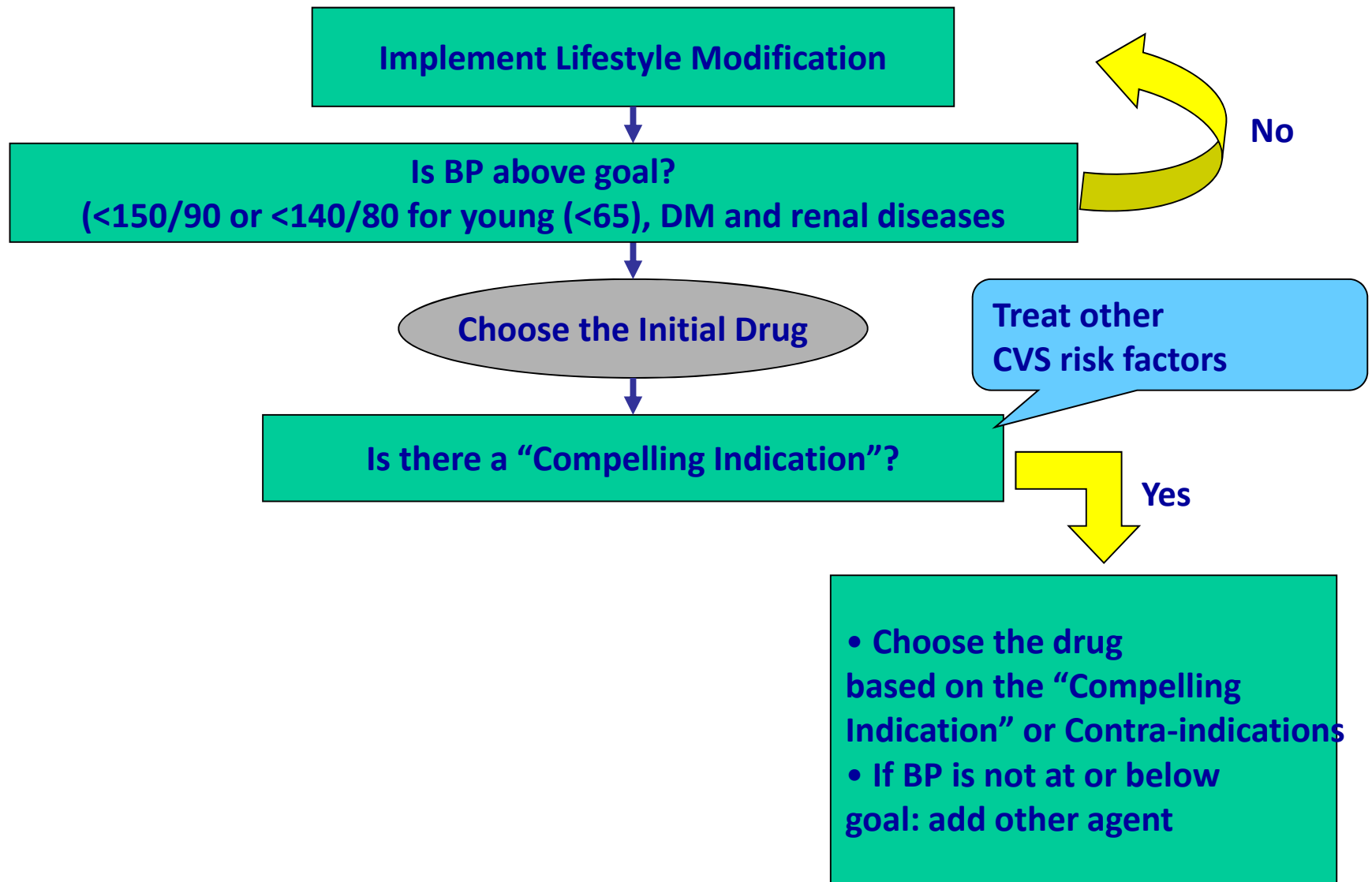
Ageing Kidneys

Blood Pressure Target Guideline

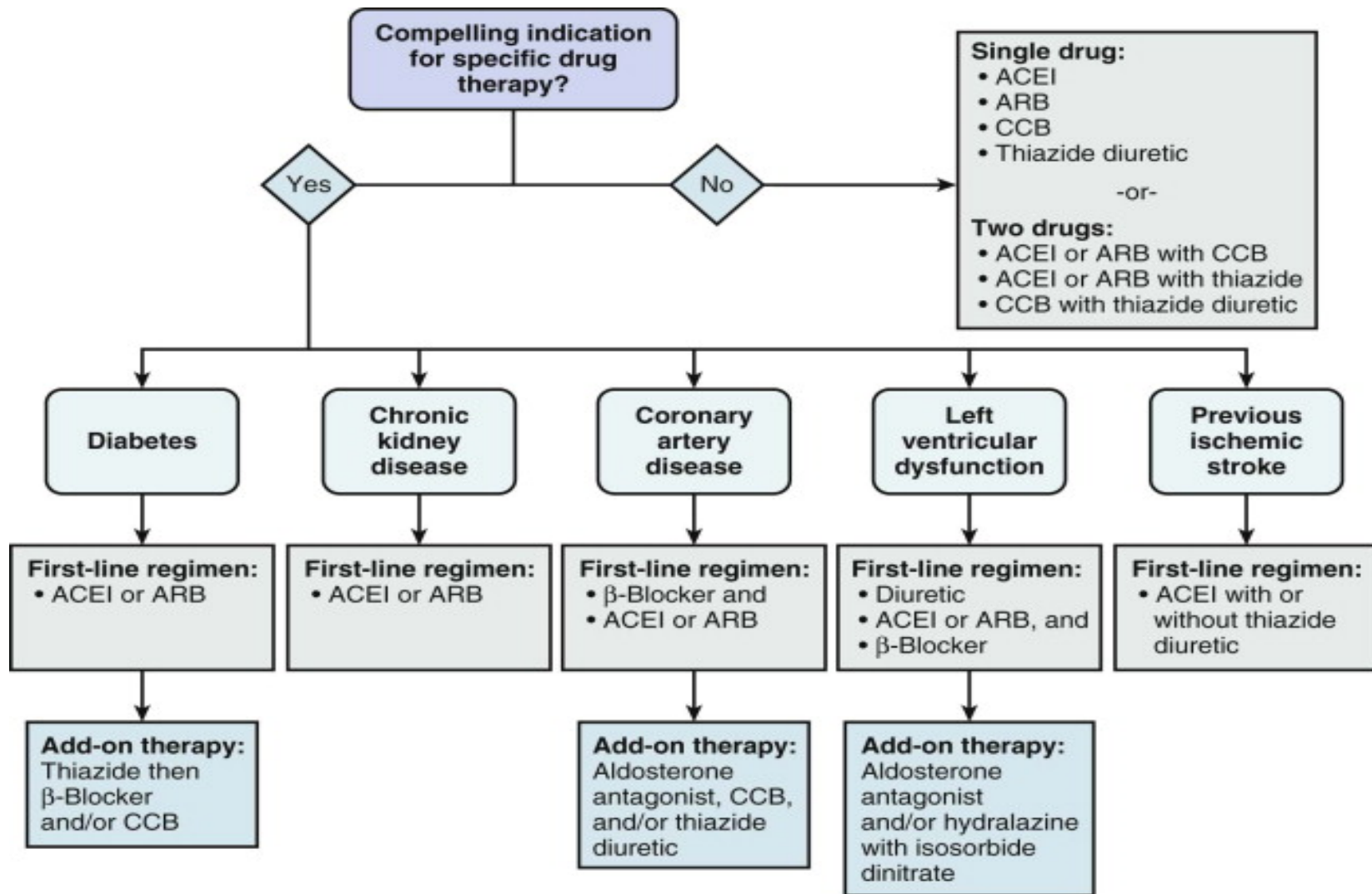
Blood Pressure Targets in International Guidelines (office BP mmHg)

Underlying Comorbidity	ACC/AHA	ESC/ESH	Canadian	JHS	NICE	ADA	KDIGO
No Comorbidity	<130/80	<130/80 ³	<140/90	<130/80	<140/90		
Older Patients	<130/80	<140/90 ¹	<120/80	<130/80	<140/90		
Established CVD	<130/80	<130/80 ³	<120/80	<130/80	<140/90		
Diabetes	<130/80	<130/80 ³	<120/80 [No Title]	<130/80	<140/90	<140/90 ²	
CKD	<130/80	<140/90 ¹	<120/80	<130/80	<140/90		<120/80
Heart Failure	<130/80	<130/80	<120/80	<130/80	<140/90		

¹If tolerated, aiming for 130/80mmHg, ²Aim for <130/80mmHg if high risk ³In younger patients, i.e. <65yrs



Compelling indications to specific drugs

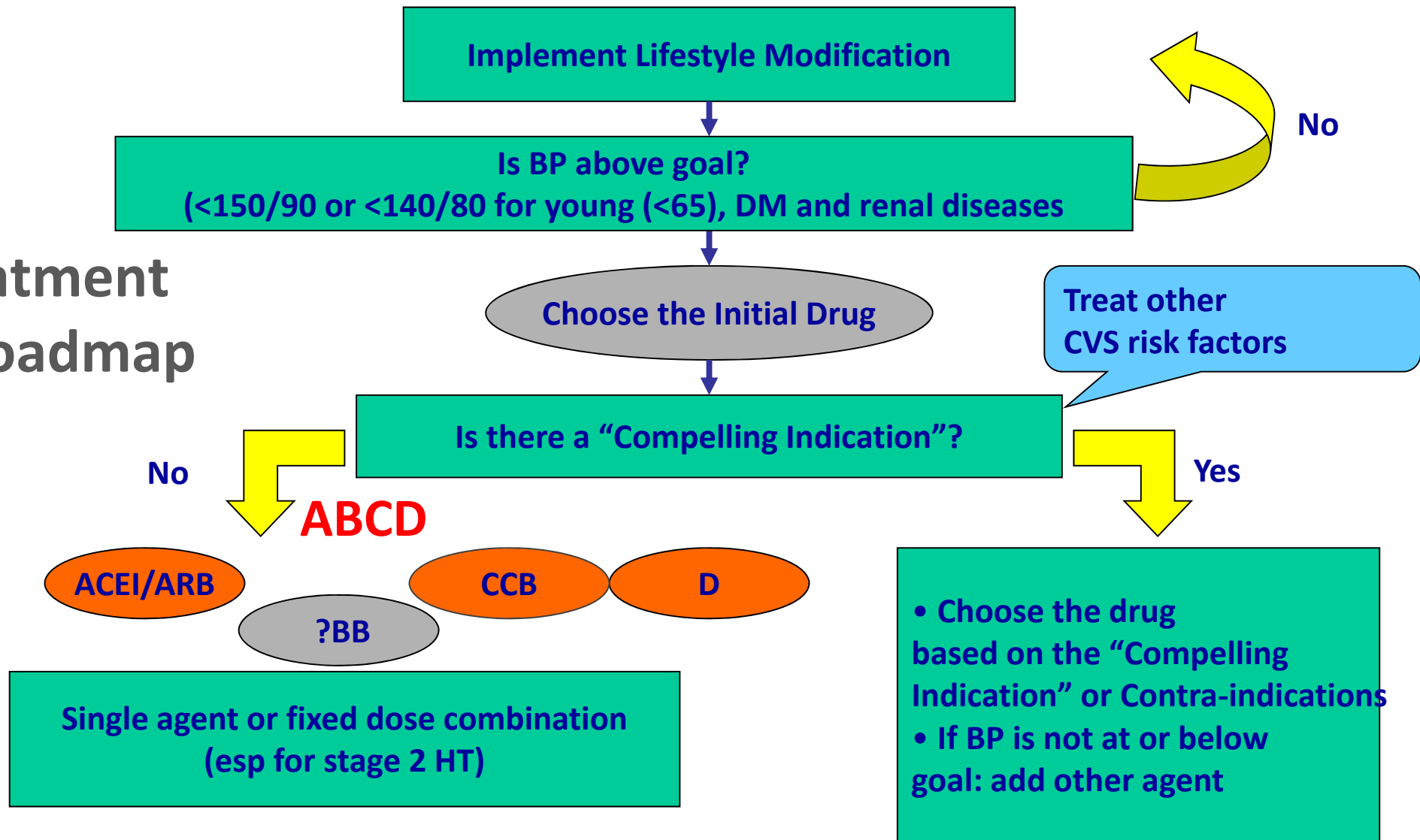


Contraindications to specific drugs

Drug	Contraindications	
	Compelling	Possible
Diuretics (thiazides/thiazide-type, e.g. chlortalidone and indapamide)	Gout	- Metabolic syndrome - Glucose intolerance - Pregnancy - Hypercalcemia - Hypokalemia
Beta-blockers	- Asthma - Any high-grade sino-atrial or atrioventricular block - Bradycardia (heart rate < 60 / min)	- Metabolic syndrome - Glucose intolerance - Athletes and physically active patients
Calcium Channel Blockers (dihydropyridines)		- Tachyarrhythmia - Heart failure (HFrEF, class III or IV) - Pre-existing severe leg oedema
Calcium Channel Blockers (verapamil, diltiazem)	- Any high-grade sino-atrial or atrioventricular block - Severe LV dysfunction (LV ejection fraction < 40%) - Bradycardia (heart rate < 60 per min)	Constipation
ACE inhibitors	- Pregnancy - Previous angioneurotic oedema - Hyperkalemia (potassium > 5.5 mmol/L) - Bilateral renal artery stenosis	Women of child-bearing potential without reliable contraception
ARBs	- Pregnancy - Hyperkalemia (potassium > 5.5 mmol/L) - Bilateral renal artery stenosis	Women of child-bearing potential without reliable contraception



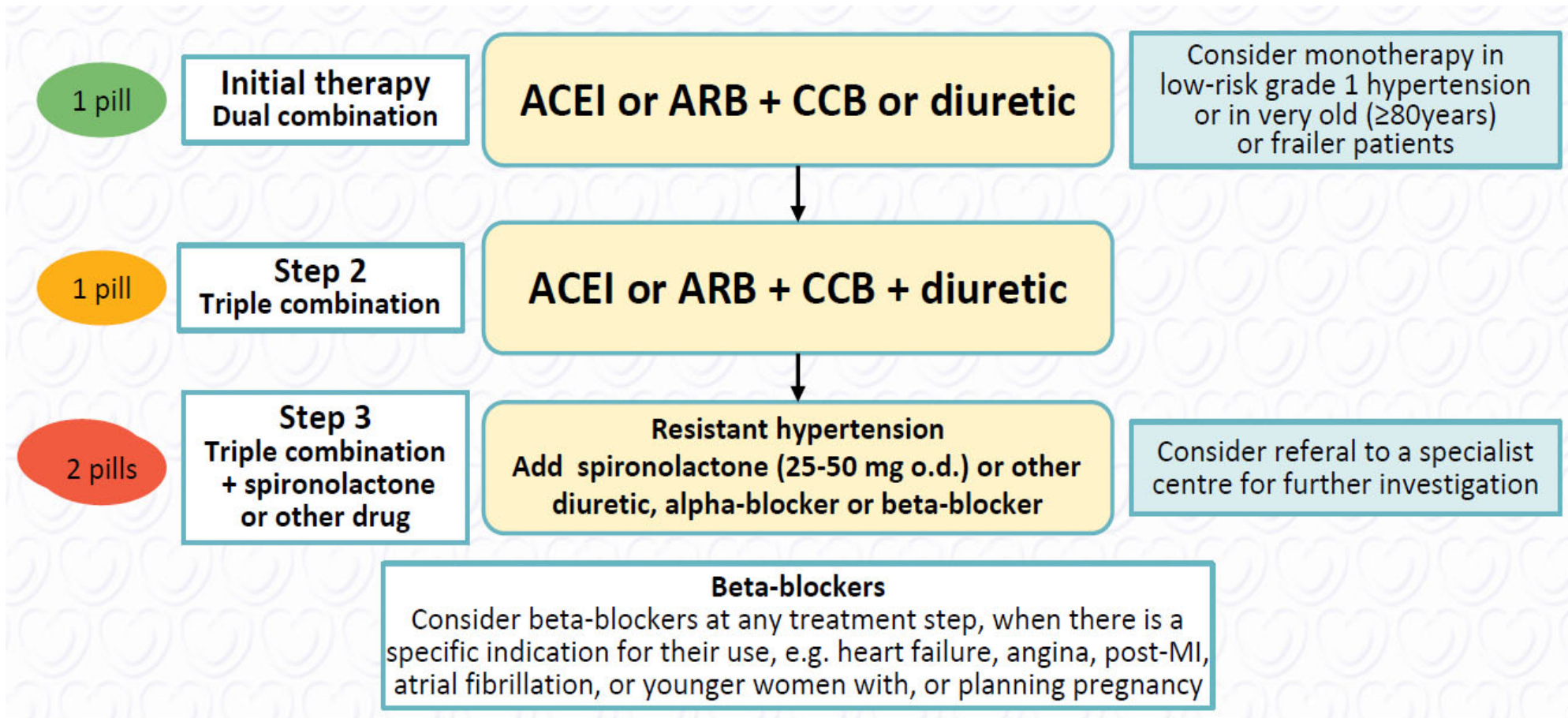
Treatment Roadmap



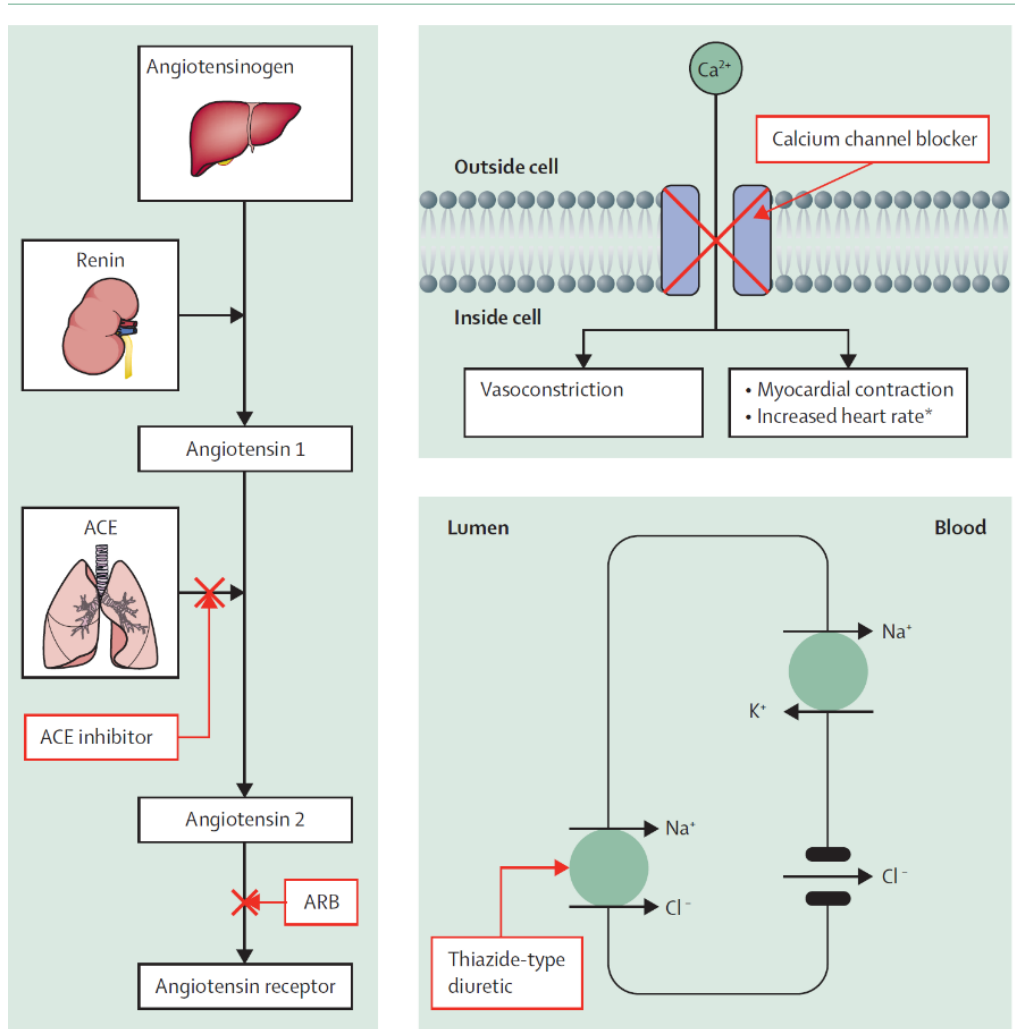
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Anti-hypertensive Rx



First-line Agents for HT



Therapy recommendations

National Institute for Health and Care Excellence⁶⁷

Patients with hypertension and diabetes or aged <55 years and not of Black African or African-Caribbean family origin (without diabetes)

- | | |
|--------|---------------|
| Step 1 | A |
| Step 2 | A plus C or D |

Patients with hypertension without diabetes and either aged ≥55 years or of Black African or African-Caribbean family origin

- | | |
|--------|---------------|
| Step 1 | C |
| Step 2 | C plus A or D |

European Society of Cardiology–European Society of Hypertension⁴⁸

- | | |
|--------|-----------------|
| Step 1 | A plus C or D |
| Step 2 | A plus C plus D |

International Society of Hypertension⁵⁰

- | | |
|--------|-----------------------|
| Step 1 | A plus C*† (low dose) |
| Step 2 | A plus C* (full dose) |
| Step 3 | A plus C* plus D |

American College of Cardiology–American Heart Association⁴⁹

- | | |
|--------|--------------|
| Step 1 | A or C or D‡ |
|--------|--------------|

A is angiotensin-converting enzyme inhibitor or angiotensin receptor blocker; C is calcium channel blocker; and D is thiazide-like or thiazide diuretic. *Dihydropyridine calcium channel blocker. †Angiotensin receptor blocker preferred in Black patients. ‡Dual therapy upfront for patients with stage 2 hypertension and blood pressure ≥20/10 mm Hg above target.

Which Combination Therapy?

CCB+ACEI/ARB

or

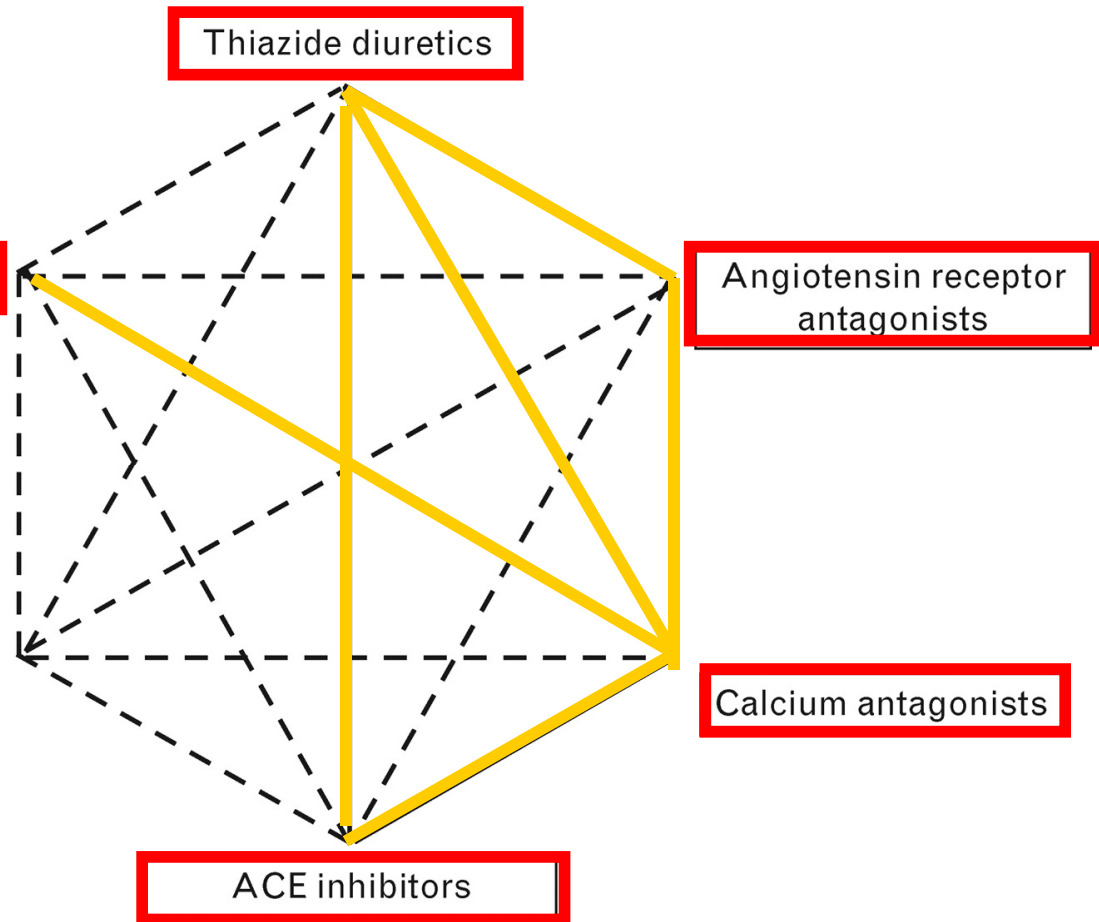
ACEI/ARB+D



CCB+ACEI/ARB+D

α -blockers

First line



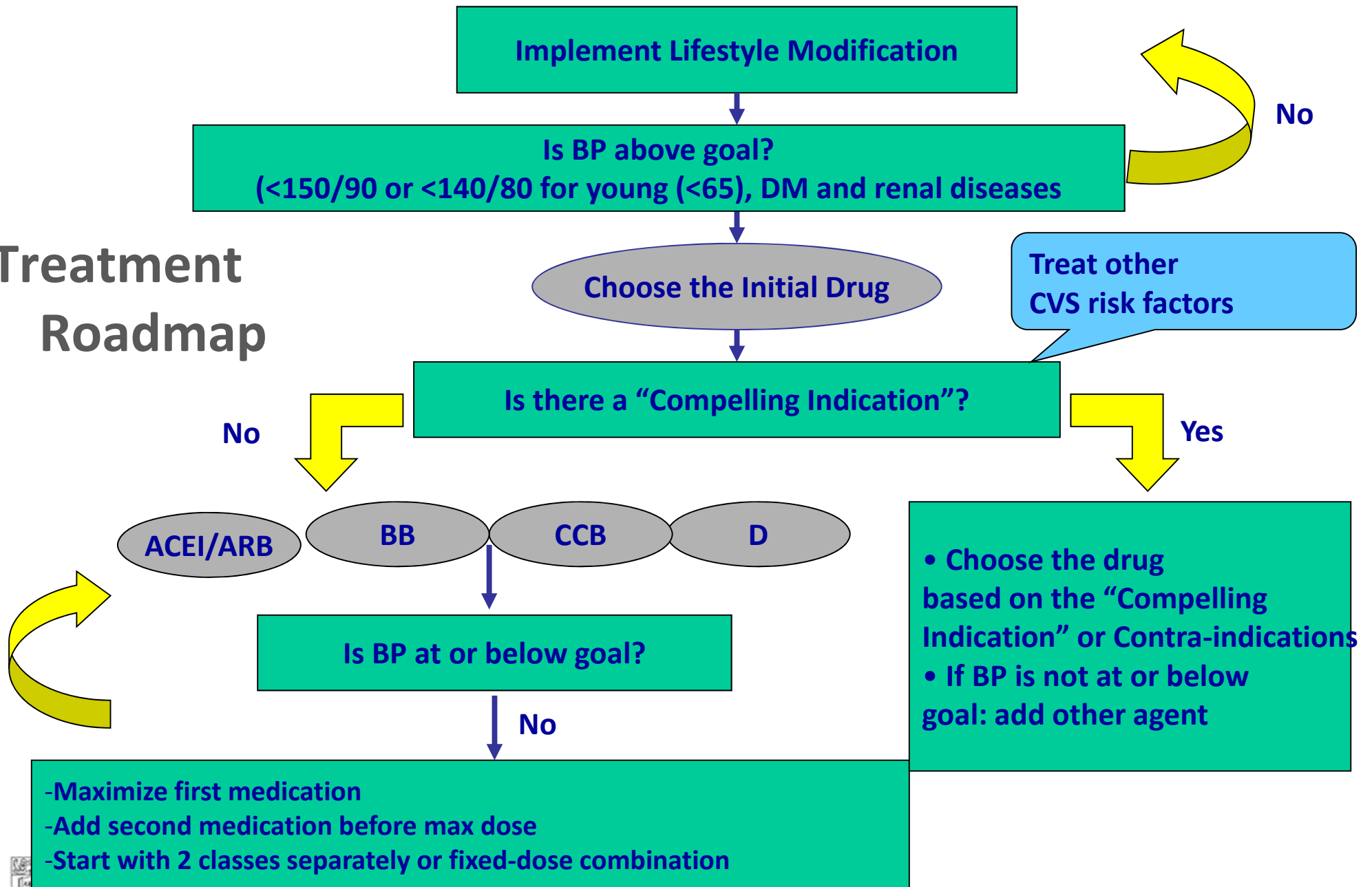
- synergic effects
- act on different mechanisms (do not combine ACEI & ARB for HT)
- counteracting adverse effects



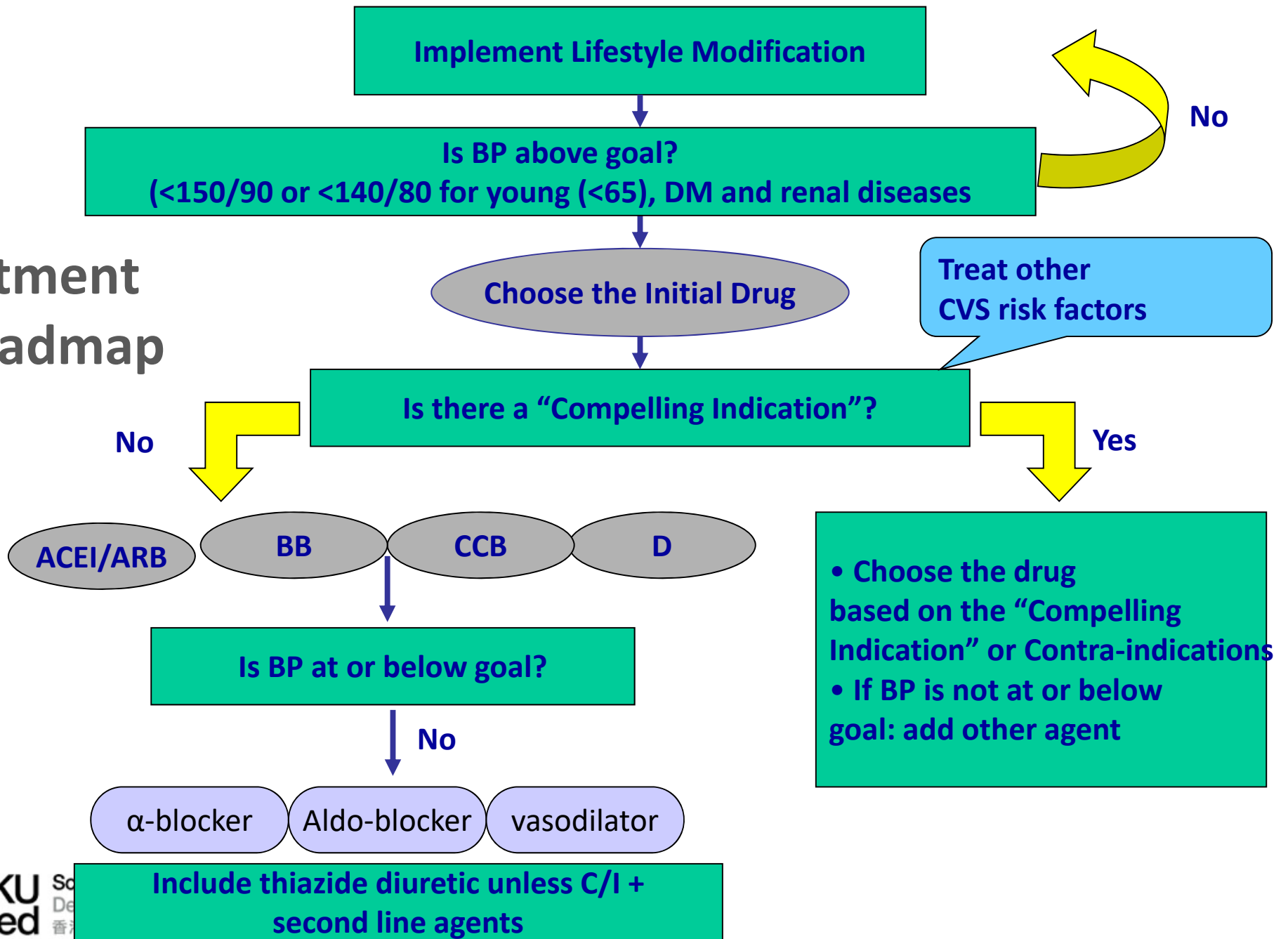
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Treatment Roadmap



Treatment Roadmap



Monitoring to specific drugs

DRUG CLASS	MONITORING PARAMETERS	ADVERSE EFFECTS
Diuretic	BP, BUN/serum creatinine, serum electrolytes (potassium, magnesium, sodium), uric acid (for thiazides especially)	Electrolyte depletion (hypokalemia, hyponatremia, hypomagnesemia), hyperkalemia with potassium-sparing agents, dehydration and orthostatic hypotension, exacerbation of gout (mostly thiazides)
Aldosterone antagonist	BP, BUN/serum creatinine, serum potassium	Hyperkalemia, especially in chronic kidney disease; dehydration and orthostatic hypotension; hyponatremia; gynecomastia with spironolactone
ACE inhibitor	BP, BUN/serum creatinine, serum potassium	Dry cough, hyperkalemia, renal failure in those with bilateral renal artery stenosis, angioedema
ARB	BP, BUN/serum creatinine, serum potassium	Hyperkalemia, renal failure in those with bilateral renal artery stenosis
Dihydropyridine CCB	BP, heart rate	Tachycardia, peripheral edema, headache, flushing, worsening GERD
Nondihydropyridine CCB	BP, heart rate	Heart block, constipation, peripheral edema, worsening GERD
β -Blocker	BP, heart rate	Exercise intolerance, fatigue, heart block, exacerbation of peripheral arterial disease, erectile dysfunction, masking signs and symptoms of hypoglycemia

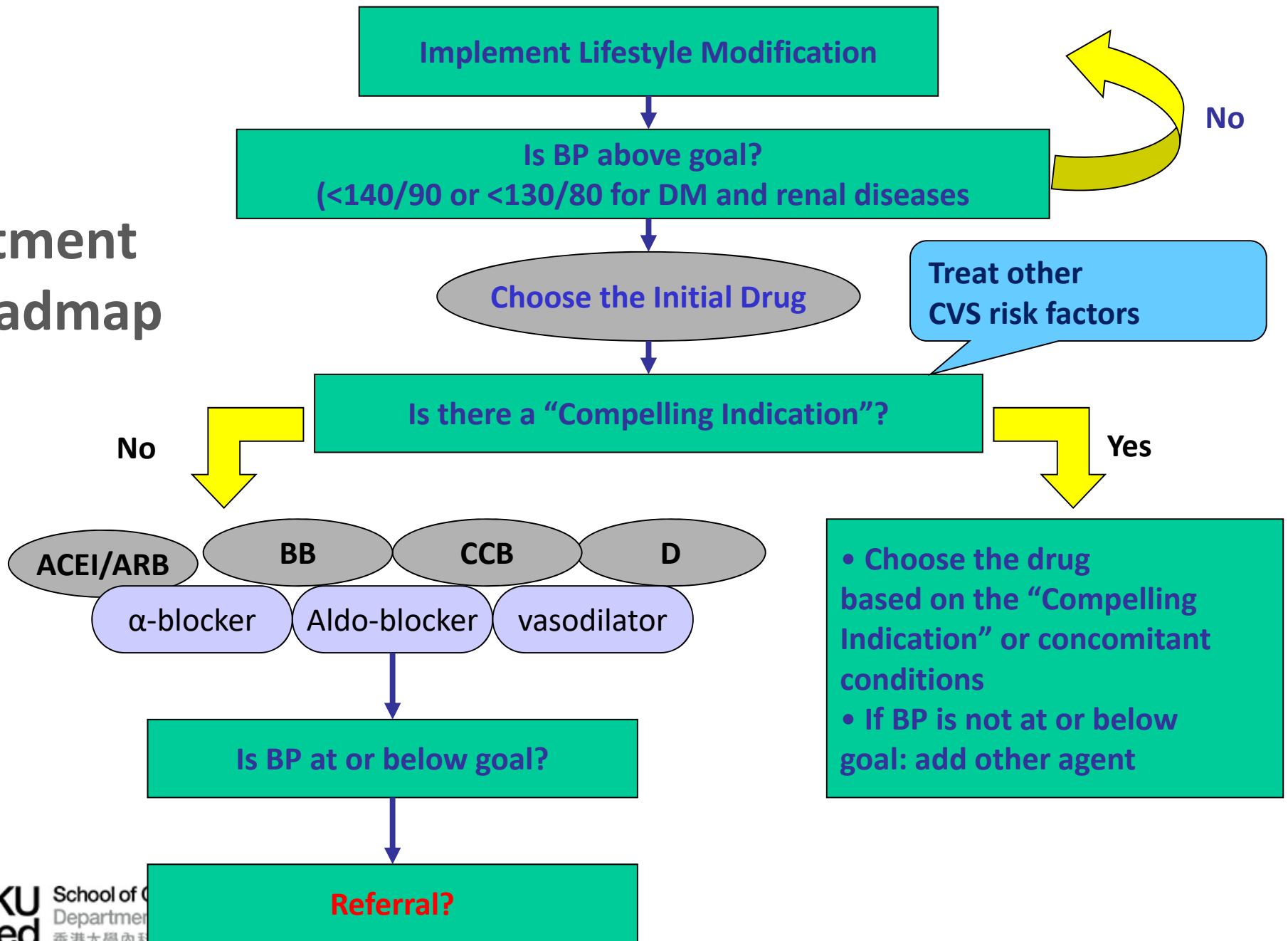


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Cardiovascular Therapeutics: A Companion to Braunwald's Heart Disease

Treatment Roadmap



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What to do if the Rx Failed?

AT-GOALs

Treatment Resistance



Exclude Pseudoresistance

- Adherence

- Timing of drug:

- Home and ambulatory BP

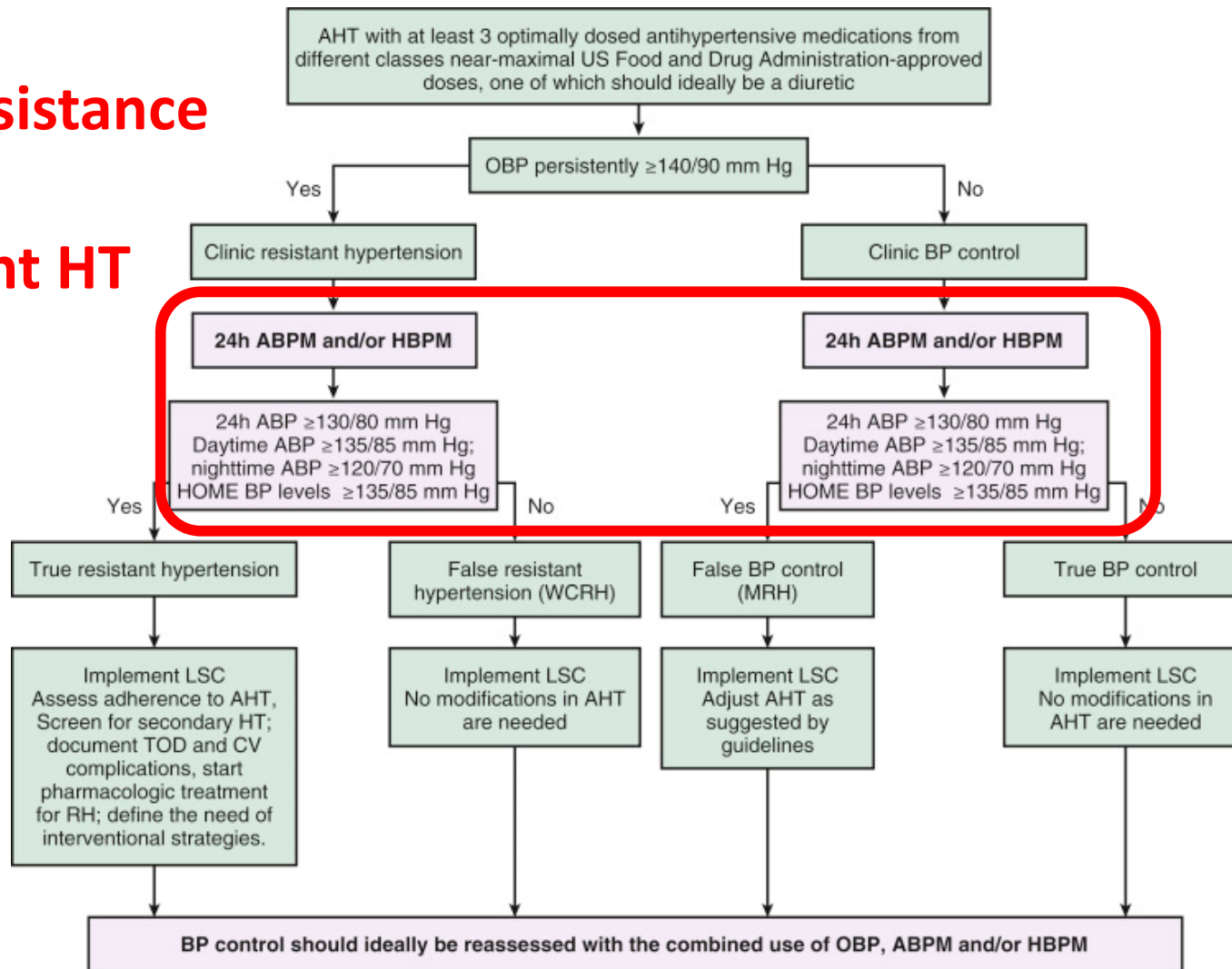


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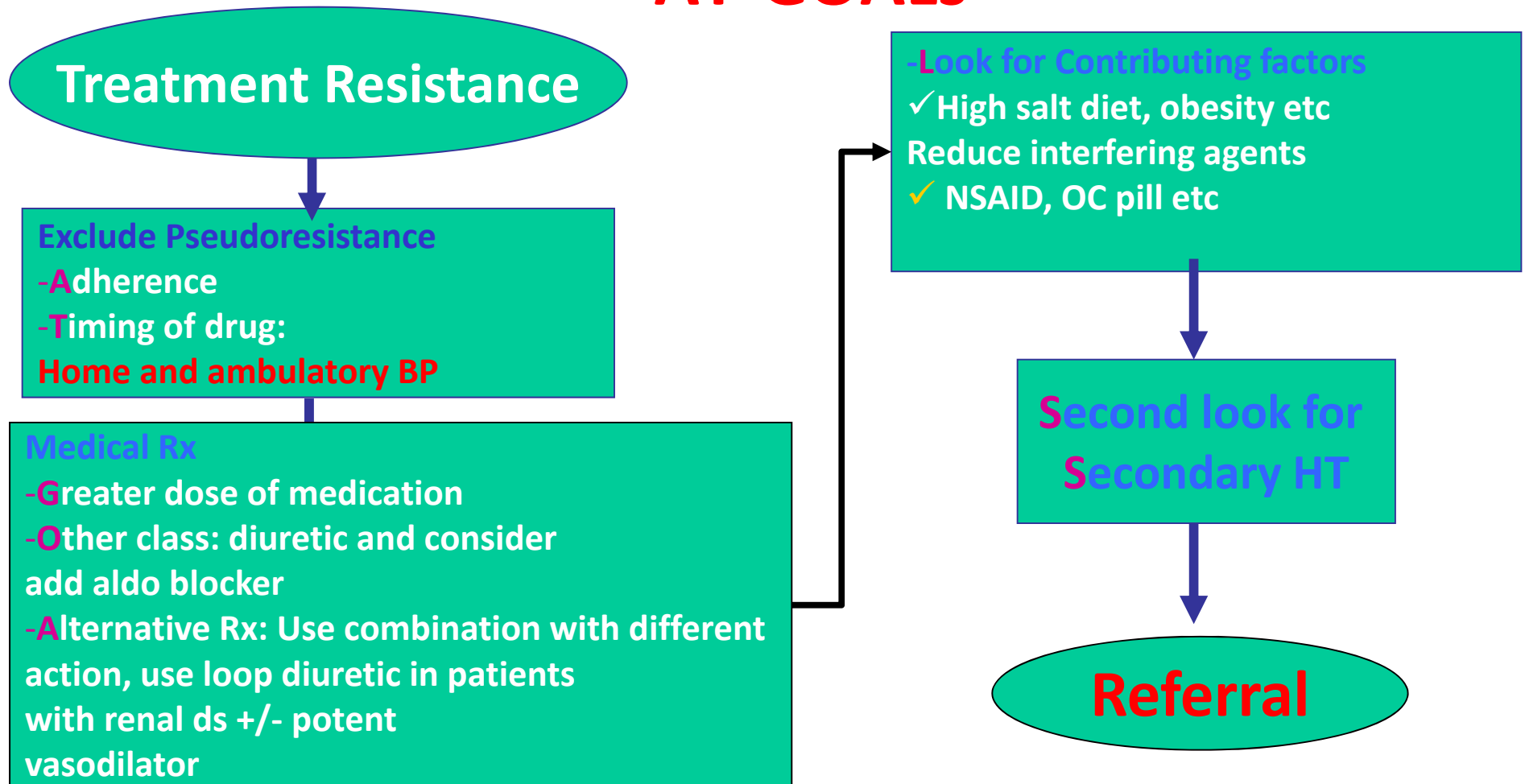
What to do if the Rx Failed?

Pseudo-resistance
Vs.
Resistant HT



What to do if the Rx Failed?

AT-GOALs



Malignant Hypertension

- < 1% of HT population, 25-50% 5 yrs mortality
- Marked BP elevation DBP >140mmHg
- Papilloedema and retinal hemorrhages and exudates
- HT encephalopathy: severe headache, vomiting, visual disturbances (including transient blindness), transient paralyses, convulsions, stupor, and coma.
- Cardiac decompensation
- Acute renal failure with oliguria



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Indications for immediate or early treatment for hypertension

Hypertensive emergency (immediate Rx-within 1 hr)

- Malignant HT*
- HT encephalopathy
- Acute heart failure
- Unstable angina/MI
- Dissecting aortic aneurysm
- Cerebral hemorrhage
- Renal failure
- Severe preclampsia
- Adrenergic crisis

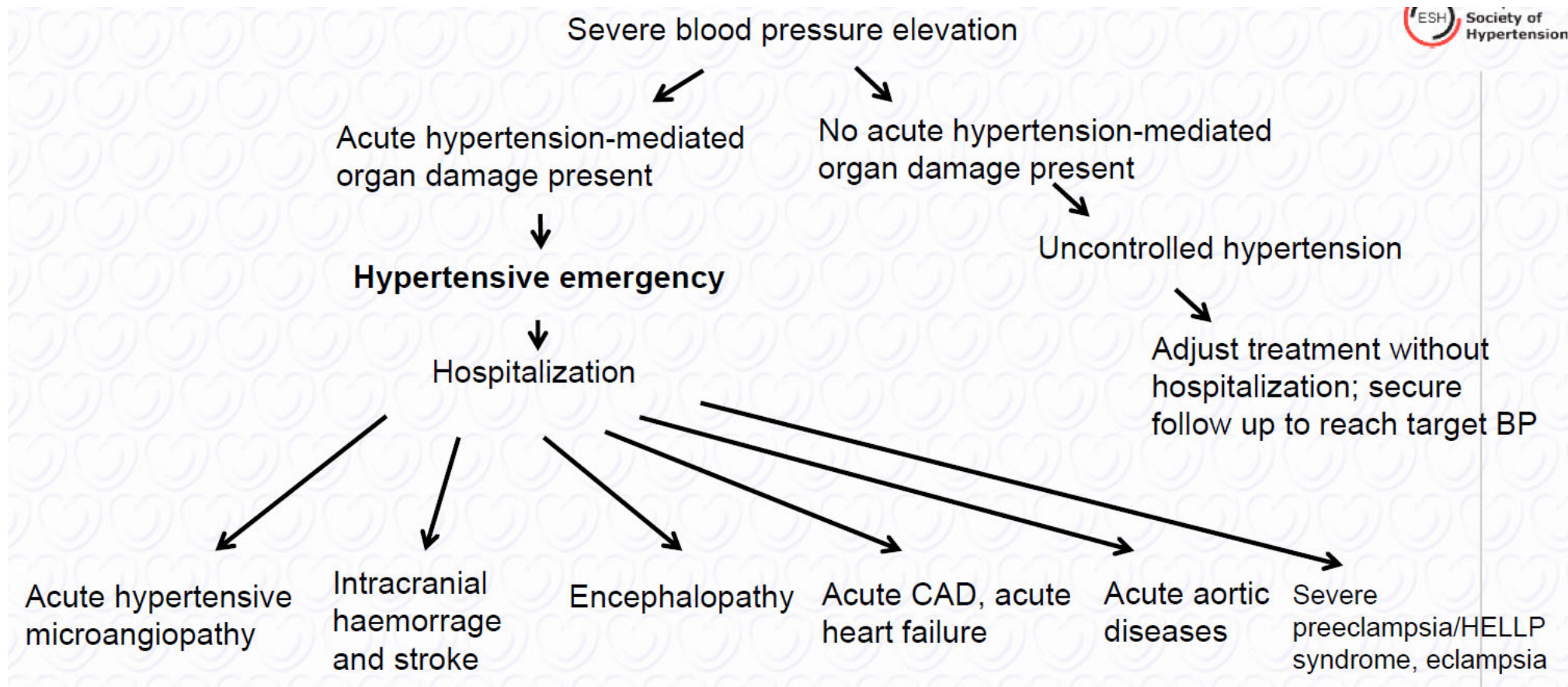
Hypertensive urgency (early Rx-within 24 hrs)

- HT with grade III or IV retinal changes
- Severe preoperative or perioperative HT

*elevated BP with encephalopathy or nephropathy or papilledema +/- microangiopathic hemolytic anemia



Hypertensive Emergencies



Indication for Emergency Reduction of BP with IV Treatment

- Hypertensive encephalopathy- malignant HT, eclampsia
 - Acute left ventricular failure due to HT
 - Dissecting aneurysm
- * Emergency reduction of BP in some conditions e.g. accelerated BP associated with acute stroke is contraindicated because of loss of cerebral autoregulation and consequent risk of cerebral infarction



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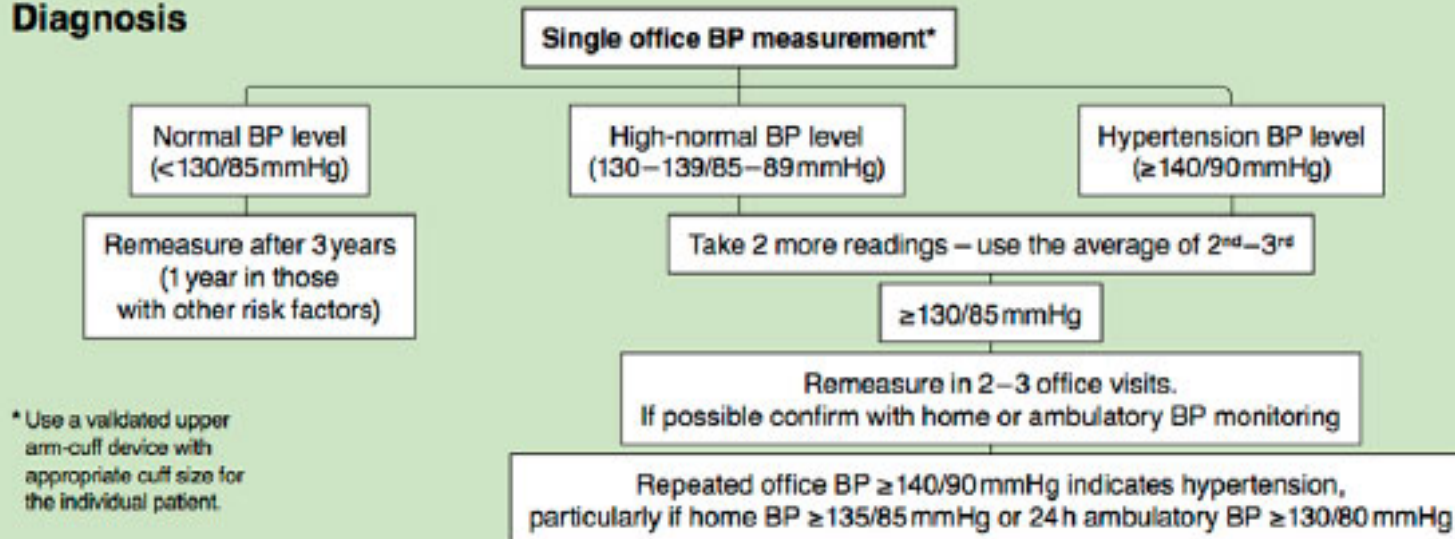
Treatment of Hypertensive Emergencies

TYPE OF EMERGENCY	TIMELINE, TARGET BLOOD PRESSURE	FIRST-LINE THERAPY	ALTERNATIVE THERAPY
Hypertensive crisis with retinopathy, microangiopathy, or acute renal insufficiency	Several hours, MAP –20% to –25%	Labetalol	Nitroprusside Nicardipine Urapidil
Hypertensive encephalopathy	Immediate, MAP –20% to –25%	Labetalol	Nicardipine Nitroprusside
Acute aortic dissection	Immediate, SBP <110 mm Hg	Nitroprusside + metoprolol	Labetalol
Acute pulmonary edema	Immediate, MAP 60-100 mm Hg	Nitroprusside with loop diuretic	Nitroglycerin Urapidil with loop diuretic
Acute coronary syndrome	Immediate, MAP 60-100 mm Hg	Nitroglycerin	Labetalol
Acute ischemic stroke and BP >220/120 mm Hg	1 hour, MAP –15%	Labetalol	Nicardipine Nitroprusside
Cerebral hemorrhage and SBP >180 mm Hg or MAP >130 mm Hg	1 hour, SBP <180 mm Hg and MAP <130 mm Hg	Labetalol	Nicardipine Nitroprusside
Acute ischemic stroke with indication for thrombolytic therapy and BP >185/110 mm Hg	1 hour, MAP less than –15%	Labetalol	Nicardipine Nitroprusside
Cocaine/XTC intoxication	Several hours, SBP <140 mm Hg	Phentolamine (after benzodiazepines)	Nitroprusside
Pheochromocytoma crisis	Immediate	Phentolamine	Nitroprusside Urapidil
Perioperative hypertension during or after CABG	Immediate	Nicardipine	Urapidil Nitroglycerin
During or after craniotomy	Immediate	Nicardipine	Labetalol
Severe preeclampsia/eclampsia	Immediate, BP <160/105 mm Hg	Labetalol (plus MgSO ₄ and oral antihypertensives)	Ketanserin Nicardipine



Summary

Diagnosis



Evaluation

History & Physical Exam

- Exclude drug-induced hypertension
- Evaluate for organ damage
- Assess total CV risk
- Search for symptoms/signs of secondary hypertension

Lab Tests

- Serum sodium, potassium & creatinine
- Lipid profile & glucose
- Urine dipstick
- 12 lead ECG

Additional Tests

- If necessary for suspected organ damage or secondary hypertension



Summary

Treatment

Grade 1 Hypertension:

140–159/90–99 mmHg

1. Start lifestyle interventions
2. Start drug treatment in:
 - High-risk patients (CVD, CKD, diabetes, organ damage, or aged 50–80 years)
 - All others with persistent BP elevation after 3–6 months of lifestyle intervention

Grade 2 Hypertension:

≥160/100 mmHg

1. Start drug treatment immediately
2. Start lifestyle intervention

Lifestyle Interventions

- Stop smoking
- Regular exercise
- Lose weight
- Salt reduction
- Healthy diet and drinks
- Lower alcohol intake

Drug Therapy Steps

Use any drugs available and include as many of those below as possible. Consider monotherapy in low-risk grade 1 hypertension and in patients aged >80 years or frail. Simplify regimen with once daily dosing and single pill combinations.

Non-Black Patients

1. Low dose ACEI/ARB* + DHP-CCB
2. Increase to full dose
3. Add thiazide/thiazide-like diuretic
4. Add spironolactone or, if not tolerated or contraindicated, amiloride, doxazosin, eplerenone, clonidine or beta-blocker

Black Patients

1. Low dose ARB* + DHP-CCB or DHP-CCB + thiazide/thiazide-like diuretic
2. Increase to full dose
3. Add diuretic or ARB /ACEI
4. Add spironolactone or, if not tolerated or contraindicated, amiloride, doxazosin, eplerenone, clonidine or beta-blocker

* No ACEI/ARB in women with or planning pregnancy

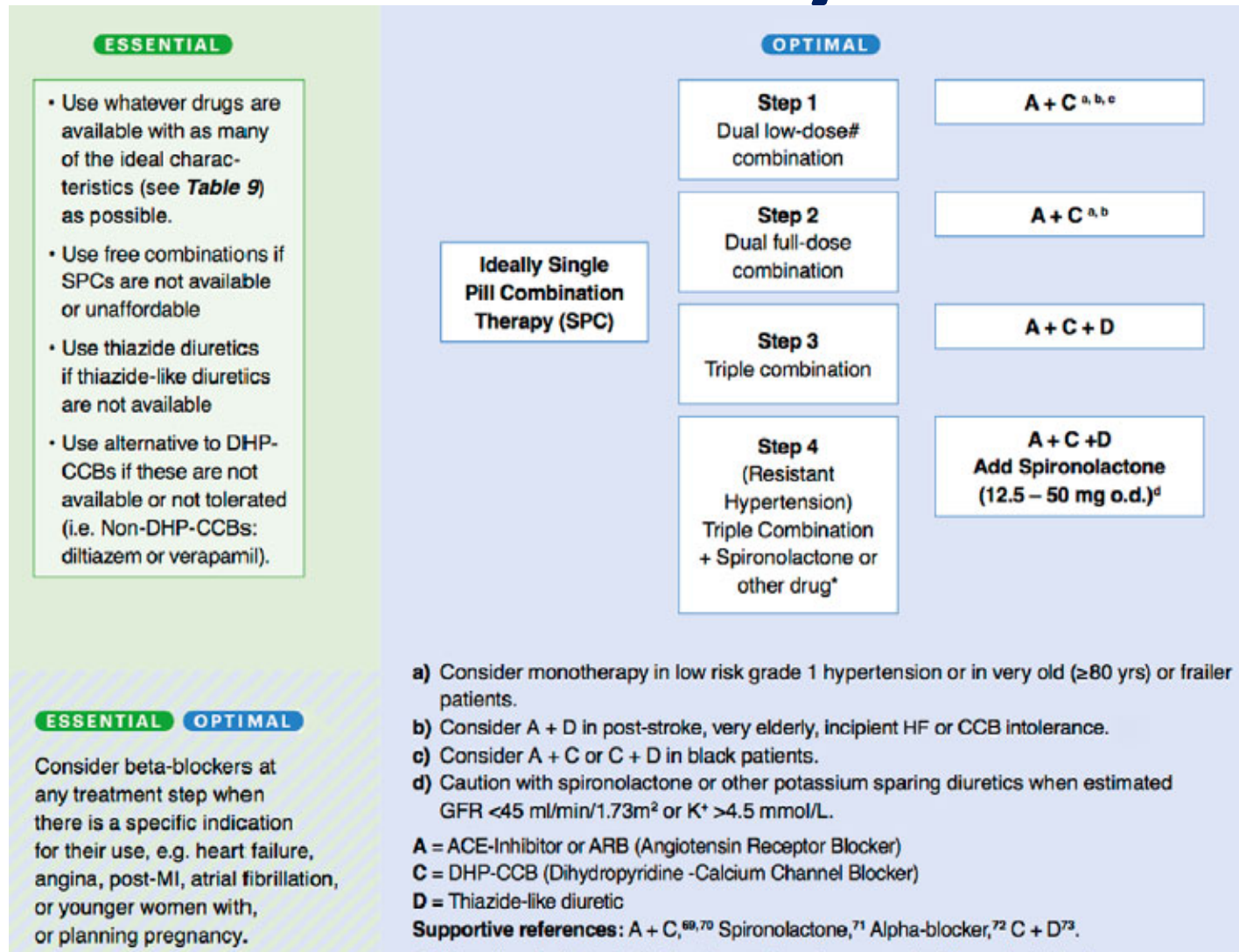


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ISH Guideline 2020. Hypertension 2020

Summary



Summary

Monitoring

Target

- Reduce BP by at least 20/10 mmHg, ideally to < 140/90 mmHg
- Individualize for elderly based on frailty

Monitor

- BP control (achieve target within 3 months)
- Adverse effects
- Long-term adherence

Referral

- If BP still uncontrolled, or other issue, refer to care provider with hypertension expertise

Recommendations: Adherence to Antihypertensive Therapy

ESSENTIAL OPTIMAL

- Evaluate adherence to antihypertensive treatment as appropriate at each visit and prior to escalation of antihypertensive treatment.
- Consider the following strategies to improve medication adherence⁸²⁻⁸⁷
 - a. reducing polypharmacy – use of single pill combinations
 - b. once-daily dosing over multiple times per day dosing
 - c. linking adherence behavior with daily habits
 - d. providing adherence feedback to patients
 - e. home BP monitoring
 - f. reminder packaging of medications
 - g. empowerment-based counseling for self-management
 - h. electronic adherence aids such as mobile phones or short messages services
 - i. multidisciplinary healthcare team approach (ie, pharmacists) to improve monitoring for adherence

OPTIMAL

- Objective indirect (ie, review of pharmacy records, pill counting, electronic monitoring devices) and direct (ie, witnessed intake of medications, biochemical detection of medications in urine or blood) are generally preferred over subjective methods to diagnose nonadherence to antihypertensive treatment.^{80,85}
- The most effective methods for management of non-adherence require complex interventions that combine counseling, self-monitoring, reinforcements and supervision.

